
Atmospheric Neutrino as a Probe of New Physics: Analysis Development in NOvA and DUNE

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Nuclear and Particle Physics Seminar

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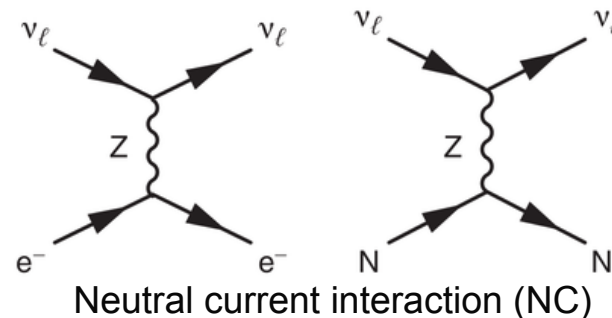
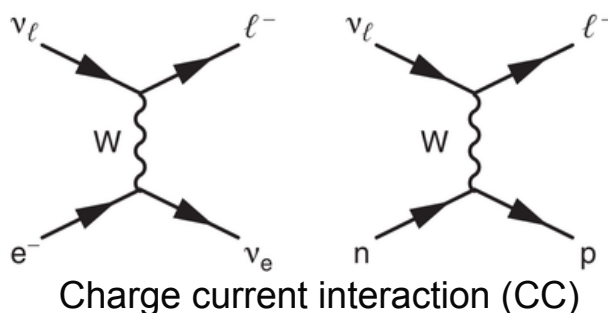
Research Questions

- Big Physics Question: Do neutrinos experience forces beyond the Standard Model when traversing matter?
- Can NOvA see atmospheric neutrinos through the cosmic ray noise? (NOvA's pioneering attempt)
- What NSI sensitivity can DUNE's atmospheric program achieve?

Neutrinos

- Neutrinos are among the most abundant particles in the universe.
- Neutrinos have no electric charge.
- Neutrinos are fermions that only interact via the weak force, and they usually pass through matter unimpeded
- Neutrino can oscillate

mass →	$\approx 2.3 \text{ MeV}/c^2$	$\approx 1.275 \text{ GeV}/c^2$	$\approx 173.07 \text{ GeV}/c^2$	0	$\approx 126 \text{ GeV}/c^2$
charge →	$2/3$	$2/3$	$2/3$	0	0
spin →	$1/2$	$1/2$	$1/2$	1	0
	u up	c charm	t top	g gluon	H Higgs boson
QUARKS	$\approx 4.8 \text{ MeV}/c^2$	$\approx 95 \text{ MeV}/c^2$	$\approx 4.18 \text{ GeV}/c^2$	0	
	$-1/3$	$-1/3$	$-1/3$	0	
	$1/2$	$1/2$	$1/2$	1	
	d down	s strange	b bottom	γ photon	
LEPTONS	$0.511 \text{ MeV}/c^2$	$105.7 \text{ MeV}/c^2$	$1.777 \text{ GeV}/c^2$	$91.2 \text{ GeV}/c^2$	
	-1	-1	-1	0	
	$1/2$	$1/2$	$1/2$	1	
	e electron	μ muon	τ tau	Z Z boson	
	$< 2.2 \text{ eV}/c^2$	$< 0.17 \text{ MeV}/c^2$	$< 15.5 \text{ MeV}/c^2$	$80.4 \text{ GeV}/c^2$	
	0	0	0	± 1	
	$1/2$	$1/2$	$1/2$	1	
	ν_e electron neutrino	ν_μ muon neutrino	ν_τ tau neutrino	W W boson	
					GAUGE BOSONS



Neutrino Oscillation: A Quantum Interference Phenomenon

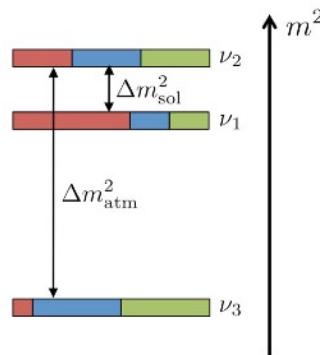
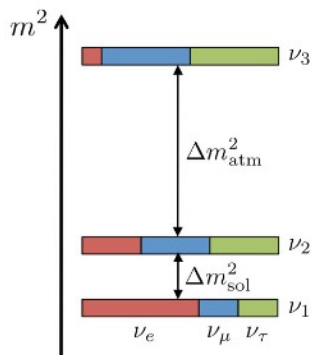
- Why oscillations happen: $|\nu_\alpha\rangle = \sum_i U_{\alpha i} |\nu_i\rangle$

$$\begin{pmatrix} |\nu_e\rangle \\ |\nu_\mu\rangle \\ |\nu_\tau\rangle \end{pmatrix} = \begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu1} & U_{\mu2} & U_{\mu3} \\ U_{\tau1} & U_{\tau2} & U_{\tau3} \end{pmatrix} \begin{pmatrix} |\nu_1\rangle \\ |\nu_2\rangle \\ |\nu_3\rangle \end{pmatrix}$$

- Oscillation probability: $P_{\nu_\alpha \rightarrow \nu_\beta} = \sum_{i,j} U_{\alpha i}^* U_{\beta i} U_{\alpha j} U_{\beta j}^* \exp\left(-i \frac{\Delta m_{ji}^2}{2E} L\right)$

Normal Ordering (NO)

Inverted Ordering (IO)



NO or IO?

Standard Oscillation

- Neutrino oscillation described by 6 free parameters:

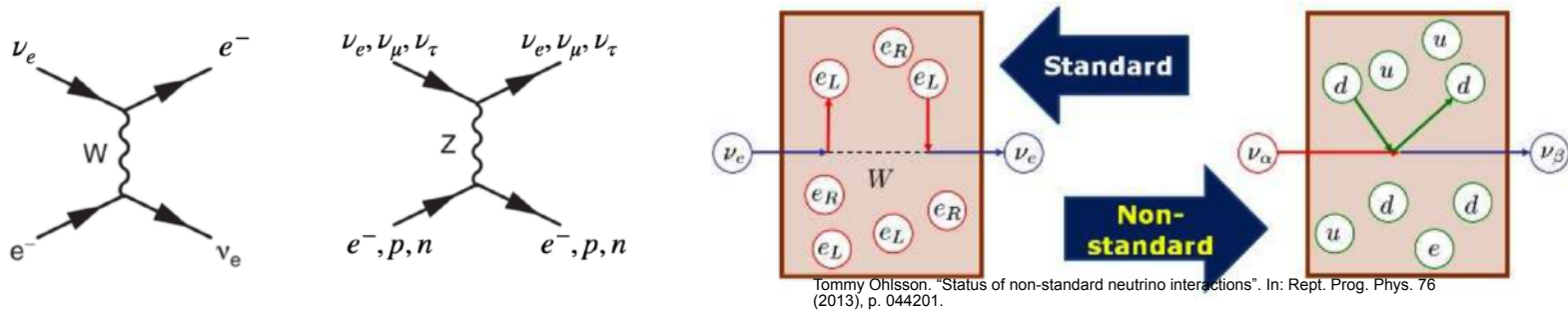
- 3 mixing angles
- 2 mass differences
- 1 CP violating phase

$$\begin{array}{c}
 \text{atm} \qquad \qquad \text{beam/reactor} \qquad \qquad \text{sol} \\
 U = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta_{CP}} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta_{CP}} & 0 & c_{13} \end{pmatrix} \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
 = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta_{CP}} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta_{CP}} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta_{CP}} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta_{CP}} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta_{CP}} & c_{23}c_{13} \end{pmatrix}
 \end{array}$$

- Experiments measure $\sin^2(2\theta_{23}) = \sin^2(2(45^\circ \pm x))$, symmetric around 45° , which octant?
- Is CP violated in lepton sector?

Nonstandard Interaction (NSI)

- In the Standard Model, only ν_e experience coherent forward (CFS) scattering from e^- via W boson
- All flavors receive the same neutral current potential, cancel out oscillation

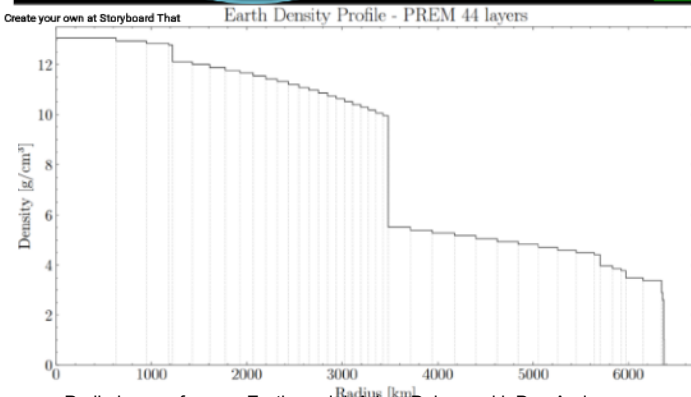
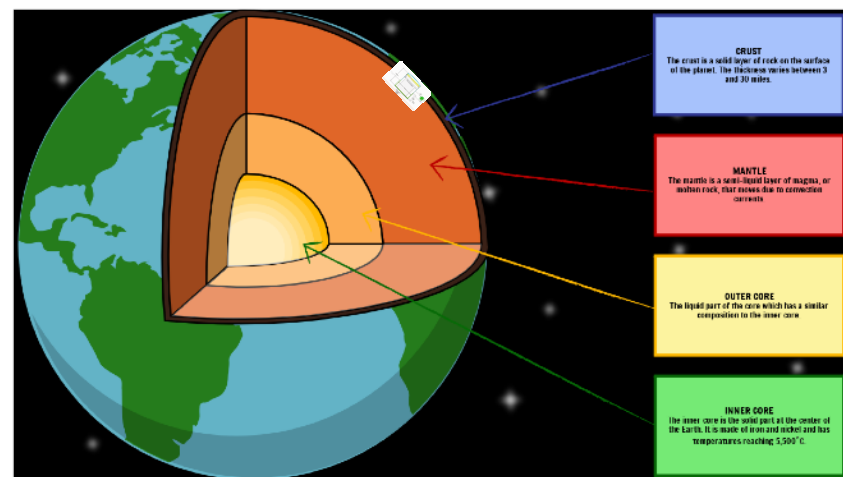


- NSI introduces new flavor dependent and flavor changing contributions to the matter potential that affect neutrino oscillation through CFS

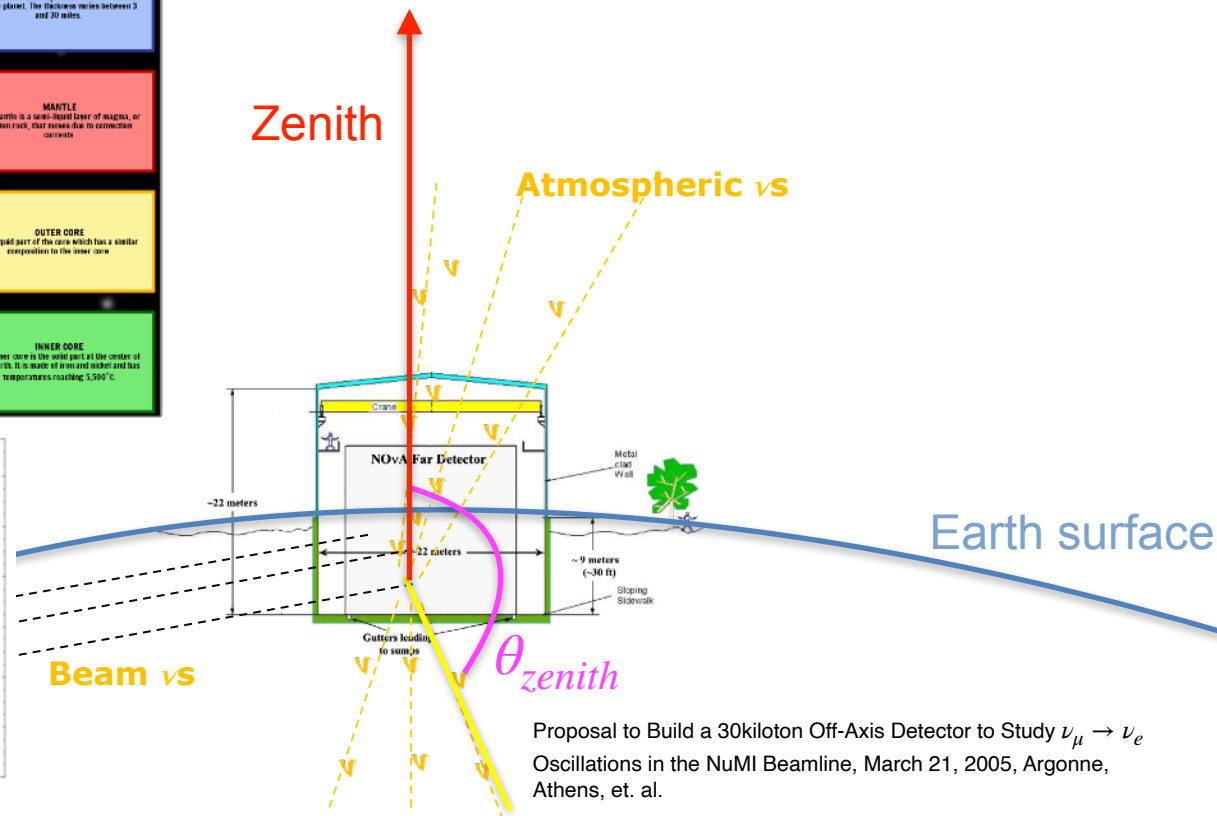
$$H_{\text{total}} = H_{\text{vac}} + V^{\text{matter}} = \frac{1}{2E} U \begin{pmatrix} 0 & 0 & 0 \\ 0 & \Delta m_{21}^2 & 0 \\ 0 & 0 & \Delta m_{31}^2 \end{pmatrix} U^\dagger + \sqrt{2} G_F \underline{N_e} \begin{pmatrix} 1 + \epsilon_{ee} & \epsilon_{e\mu} & \epsilon_{e\tau} \\ \epsilon_{e\mu}^* & \epsilon_{\mu\mu} & \epsilon_{\mu\tau} \\ \epsilon_{e\tau}^* & \epsilon_{\mu\tau}^* & \epsilon_{\tau\tau} \end{pmatrix}$$

Atmospheric Neutrinos: A Unique Laboratory for New Physics

- What is atmospheric neutrino advantage? L/E span 2 orders of magnitude!
- Direction and density
- Complementary to Beam: Probes different systematics and parameter space, helping break degeneracy that appear in beam only analyses.



Preliminary reference Earth model, Adam Dziewonski, Don Anderson

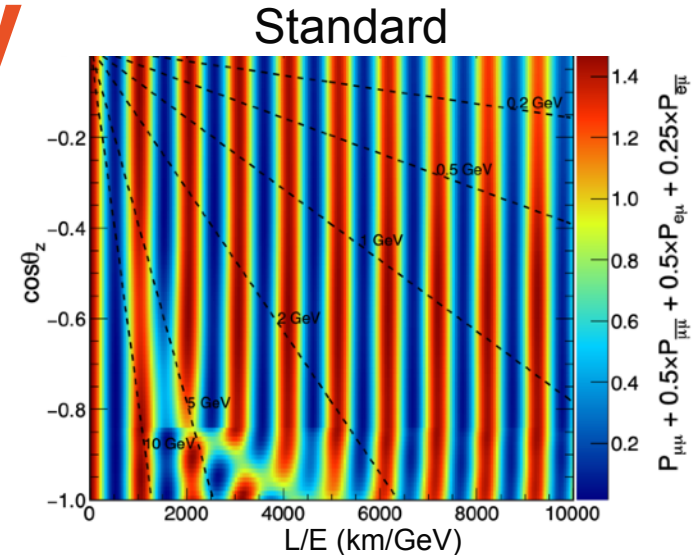


Why Atmospheric Neutrino Matters for NOvA

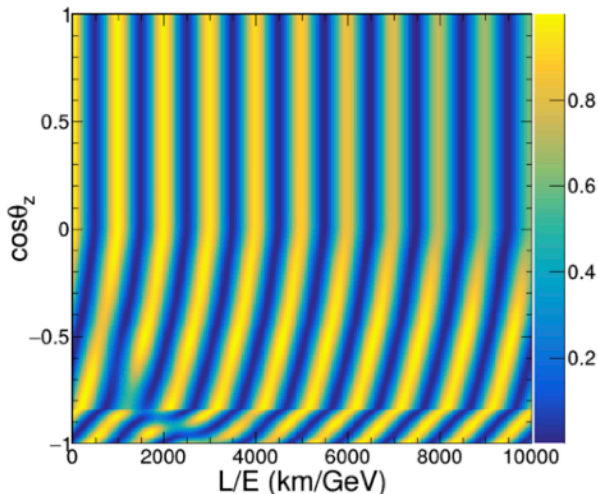
- NOvA has been collecting atmospheric data for nearly a decade, yet no analysis has been completed.
- This created a real physics opportunity: no one had quantified whether atmospheric neutrinos at a surface detector like NOvA were actually usable.
- My goal was to answer this open question: Can NOvA realistically extract atmospheric neutrino signals with useful purity and resolution?
- My work provides the first complete attempt to answer this.

NSI phenomenology

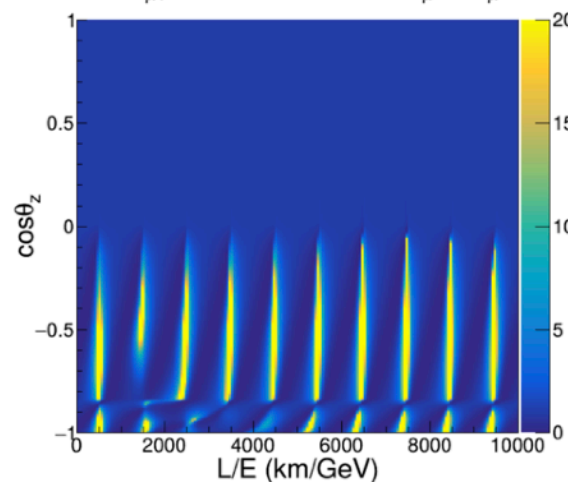
- ν_μ survival dominates ($\sim 70\%$) atmospheric statistics and most sensitive to $\mu - \tau$ sector NSI
- $\varepsilon_{\mu\tau}$ show strong distinction at large L/E



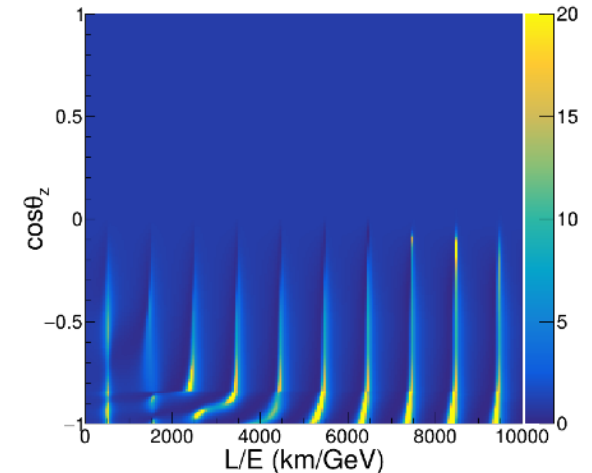
$\nu_\mu \rightarrow \nu_\mu$ With $\varepsilon_{\mu\tau} = 0.25$



$\varepsilon_{\mu\tau}$ ratio to baseline for $\nu_\mu \rightarrow \nu_\mu$

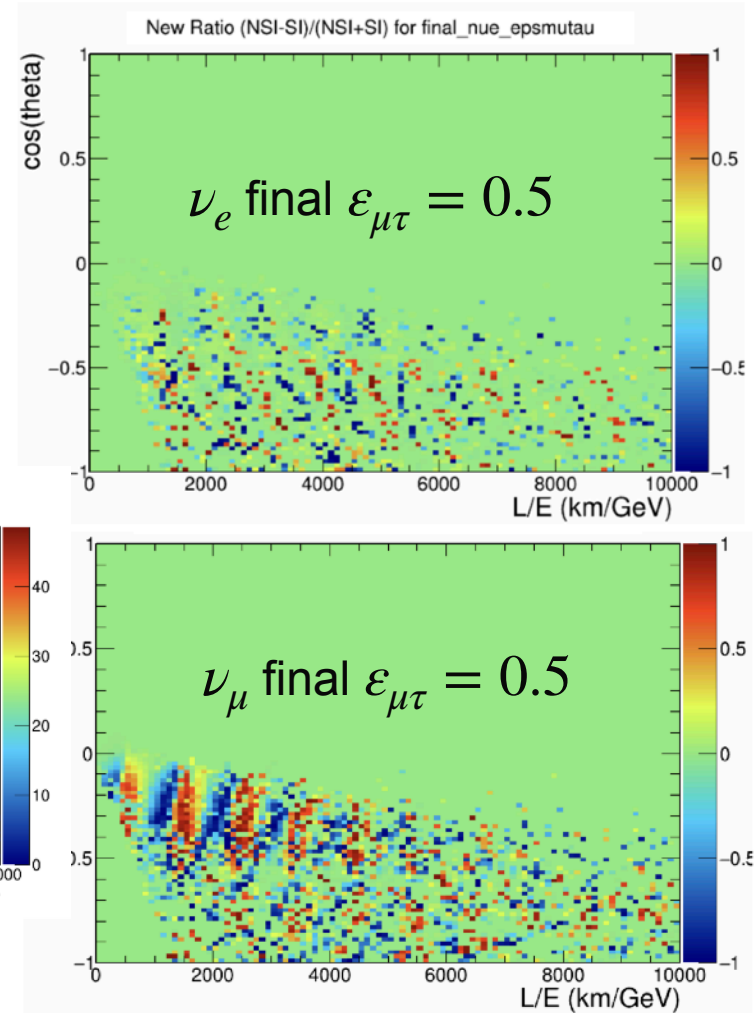
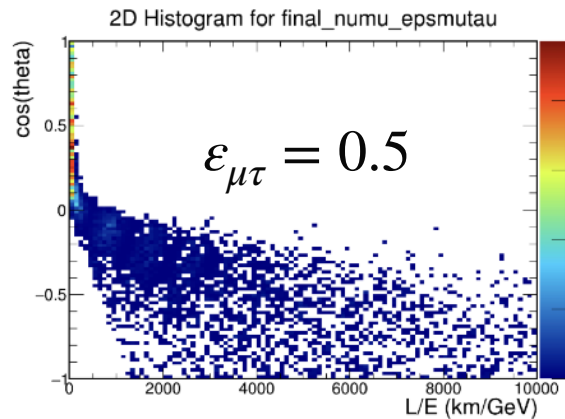
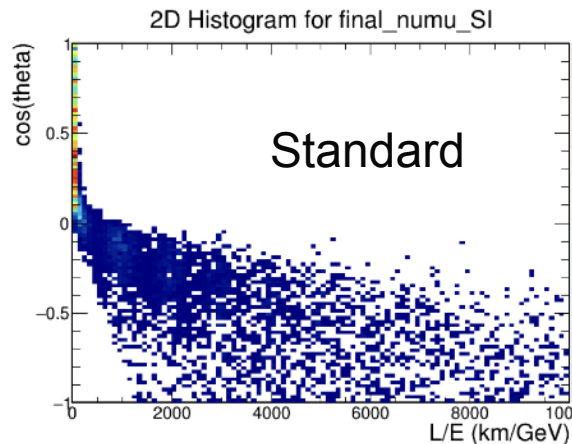


$\varepsilon_{e\mu}$ ratio to baseline for $\nu_\mu \rightarrow \nu_\mu$



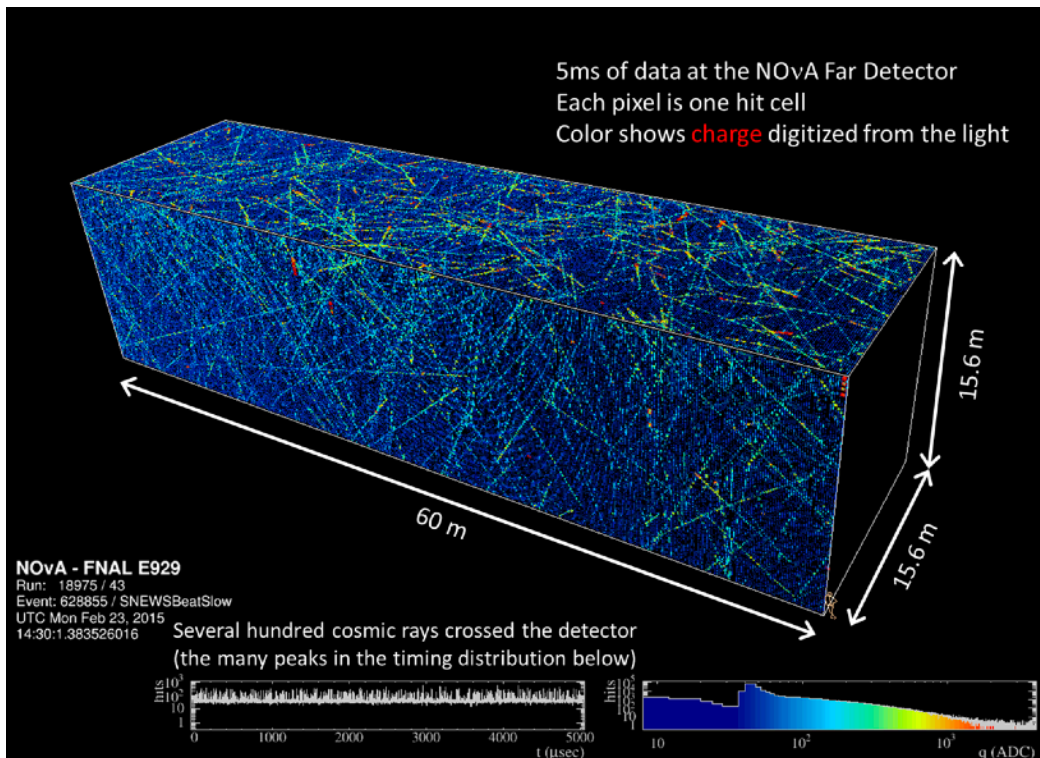
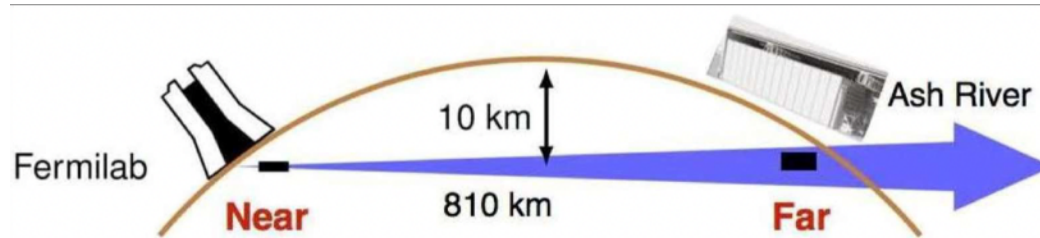
NSI Signals in NOvA Monte Carlo

- Apply oscillogram to NOvA Monte Carlo simulation
- NSI modifies both final state distributions
- Can identify NSI signal from NOvA FD MC
- Challenge: Can I extract this subtle signal from cosmic ray background?



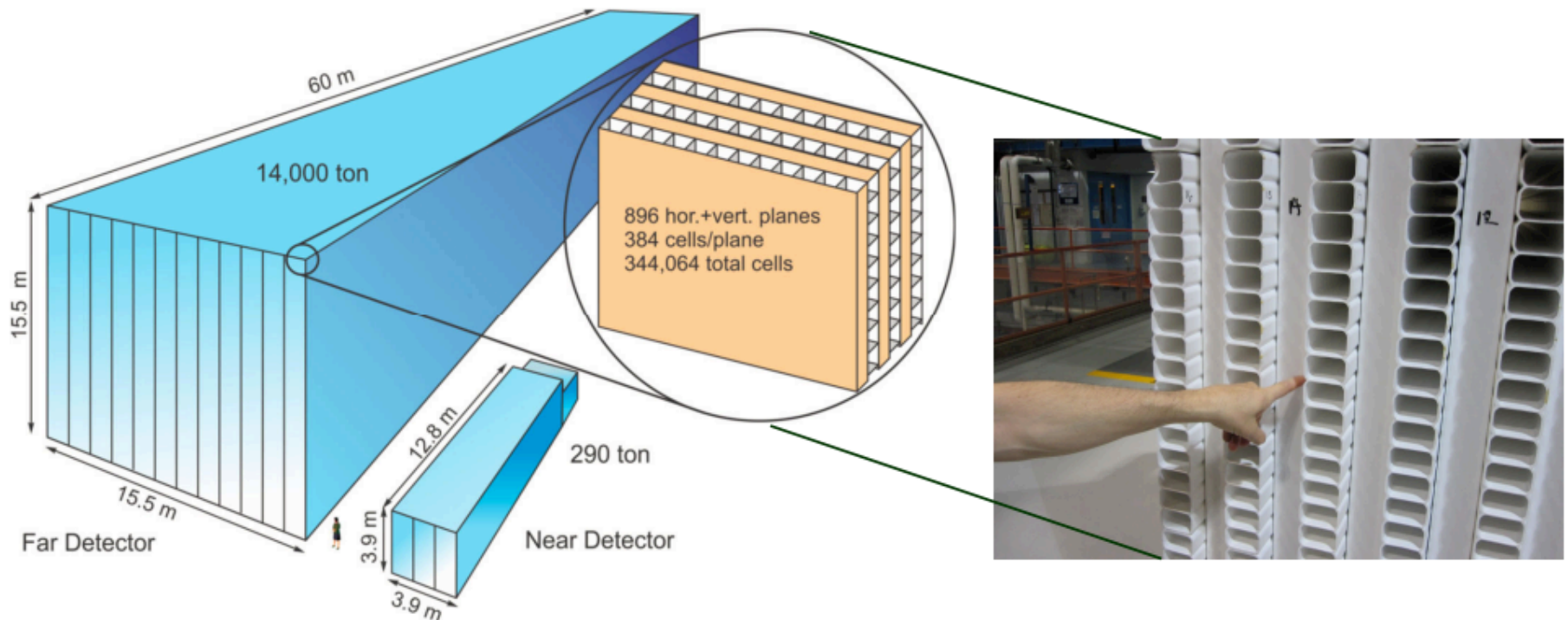
(b) With NSI: $\epsilon_{\mu\tau} = 0.5$

NuMI Off-Axis ν_e Appearance Experiment (NOvA)



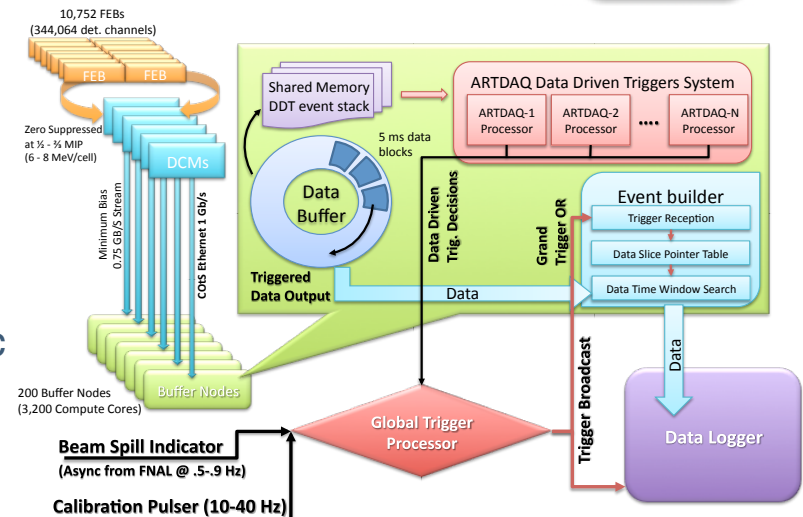
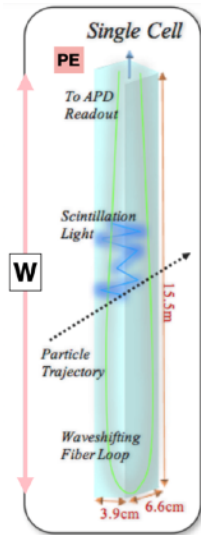
NOvA Far Detector Overview

1. It contains 14 kilotons mineral oil tracking calorimeter.
2. Constructed of horizontal & vertical planes, each with 12 extrusion modules. Each module contains 32 PVC cells.
3. Operating since 2014.



Detector Design Continued

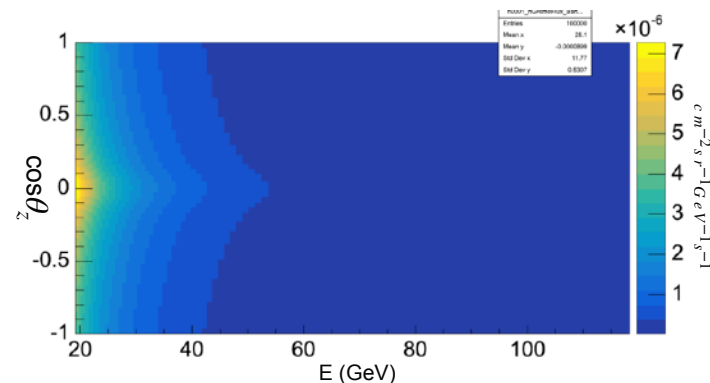
- Liquid scintillator tracking calorimeter:
 - Alternating X-Y plane enable 3D reconstruction
 - dE/dx profiles distinguish electrons from muons
- Due to huge amount of data flow ($\sim 1.2\text{GB/s}$), a trigger algorithm decide whether to keep or discard the physics information.
- Global Trigger uses 3 signals to make storage decision:
 - Beam Spill Indicator
 - Calibration Pulser
 - Data Driven Trigger (DDT)
- To understand how well we can separate signal and background, need full atmospheric neutrino simulation to study trigger efficiency



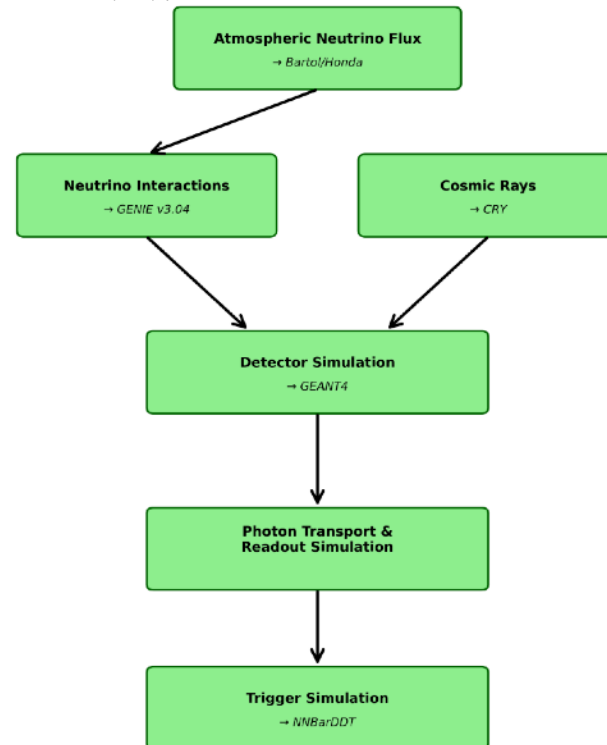
M. Tamsett A.J.Norman. DDT blessed plots Neutrino. Tech. rep. NOvA Internal Document, DocDB 11471-v1, 2014.

NOvA Simulation

- GENIE simulate neutrino and interaction by:
 - Calculate atmospheric neutrino production from cosmic ray interaction with Earth atmosphere
 - Model neutrino interaction
 - Generate interaction vertex and final state interaction
- Cosmic overlay: real 8 Hz cosmic data on neutrino for realistic background
- GEANT4 detector simulation: detector response, scintillation light, fiber transport, APD readout
- Data Driven Trigger (DDT) simulation (To be discussed next)

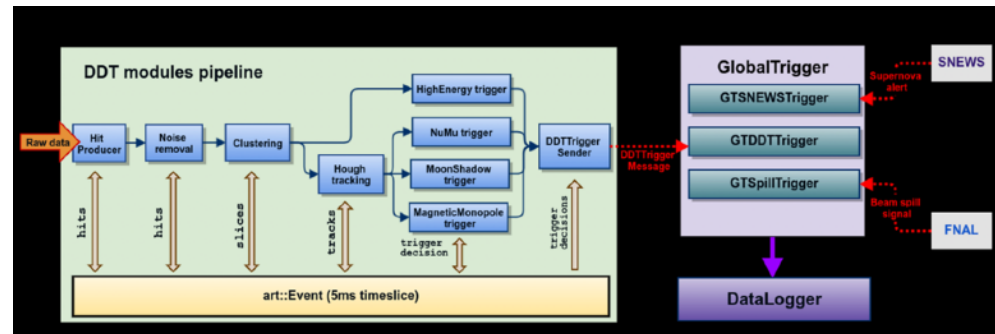


G. D. Barr and et al. "Three-dimensional calculation of atmospheric neutrinos". In: Phys. Rev. D 70 (2004), p. 023006.

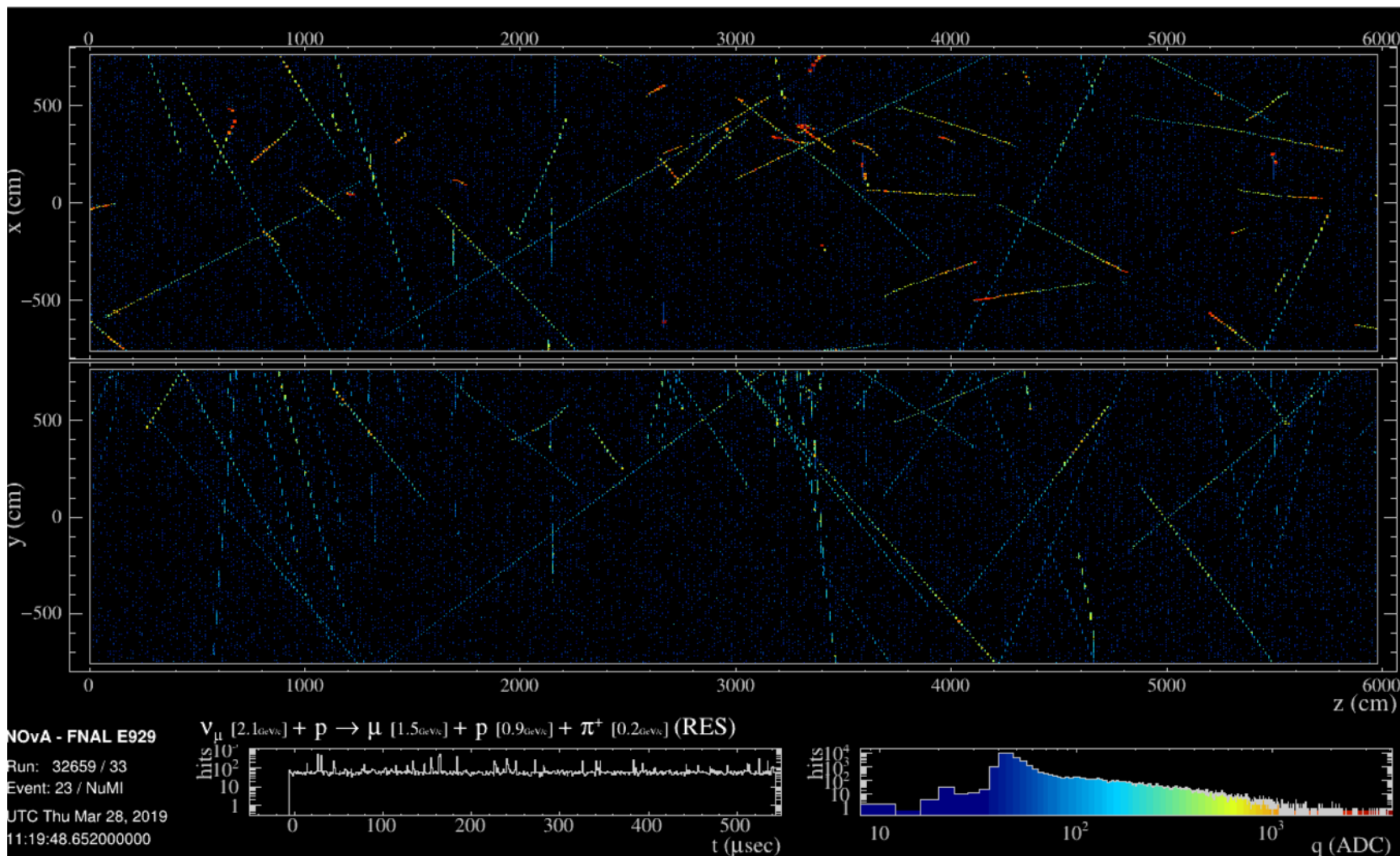


Data Driven Trigger (DDT)

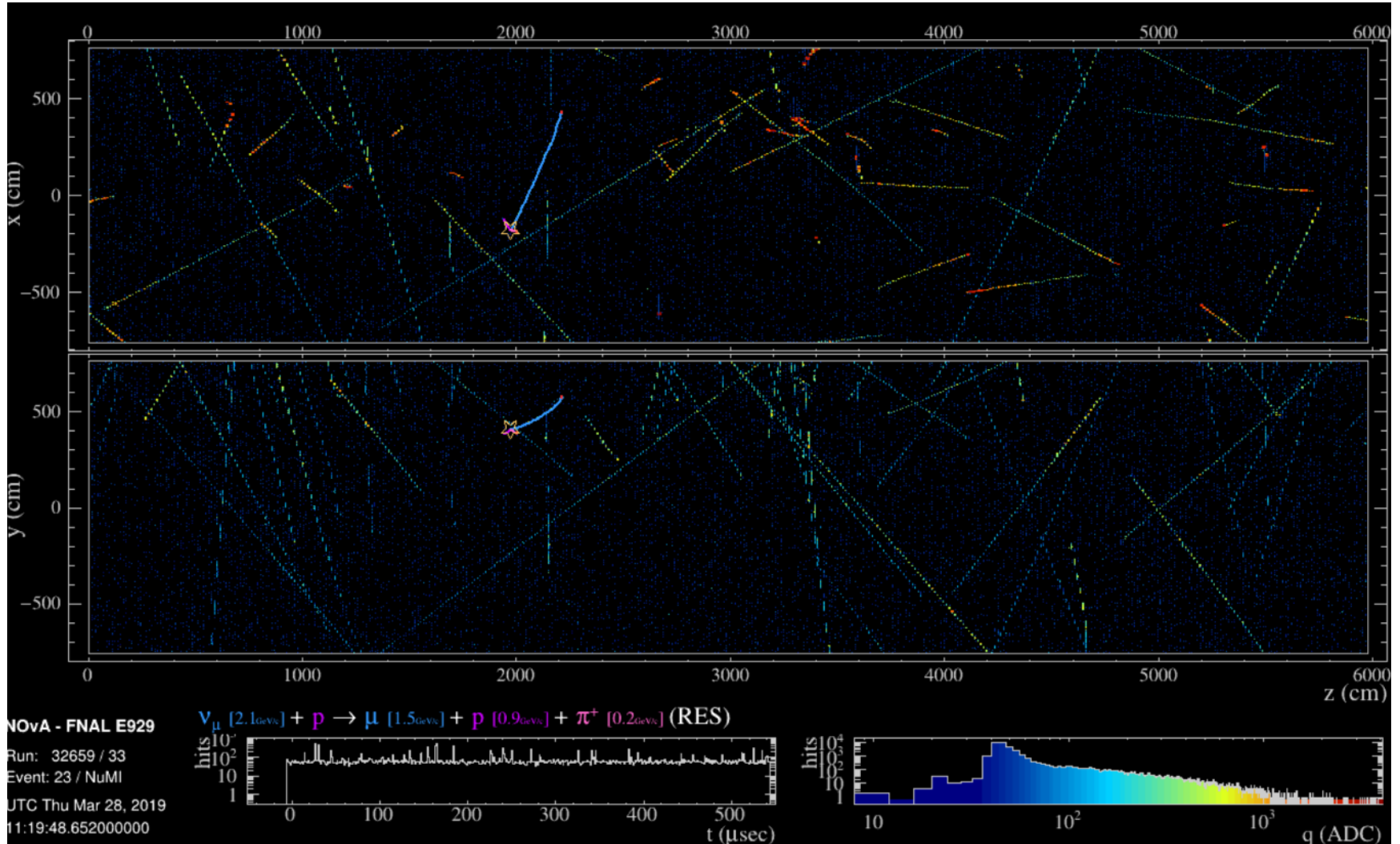
- Question: how many triggered events are neutrinos?
- Why neutron anti-neutron trigger (NNBar) ?
 - UpMu trigger: < 1% efficiency
 - NNBar topology match atmospheric ν events
- Monte Carlo simulation to validate NNBar trigger performance
 - 11.09% signal efficiency
 - $\nu_e CC$: 2748
 - $\nu_\mu CC$: 2642
 - 0.81% cosmic contamination
 - 6,000 ν_e like cosmic background
 - 6×10^6 ν_μ like background



Event Display

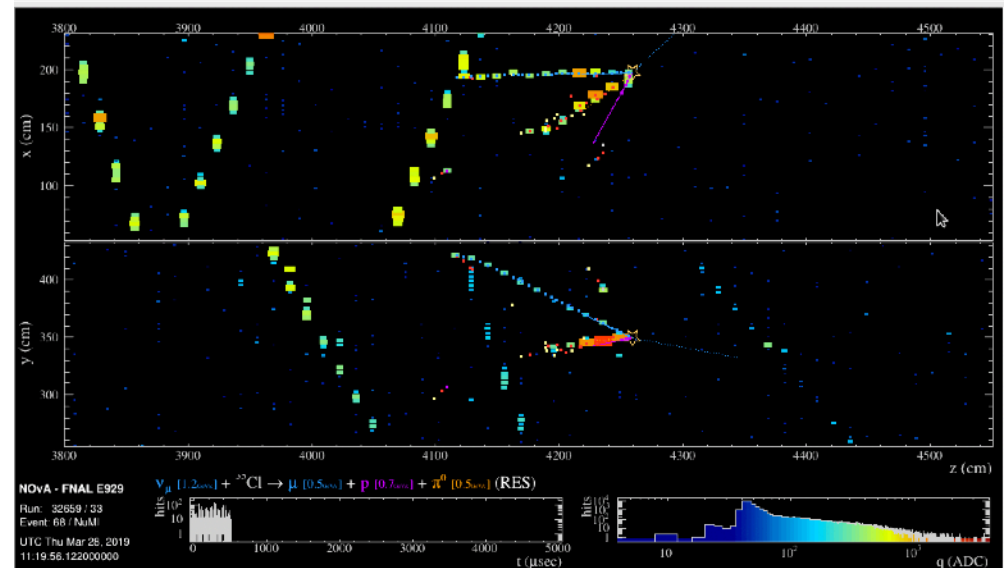


Event Display



Reconstruction

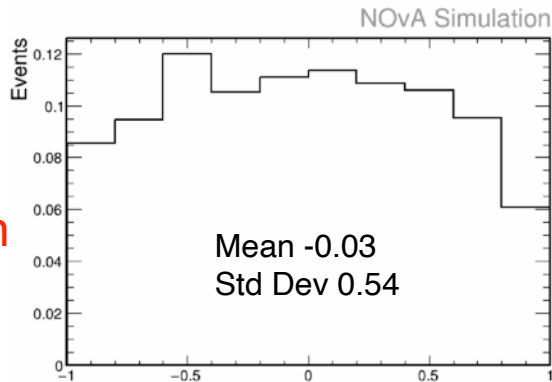
- Goal: extract neutrino energy, direction and particle ID from detector hits
- Reconstruction chain:
 - Slicer: cluster hits by space-time proximity
 - Vertex finding: locate intersection from tracks and elastic arms
 - Prong identification: trajectory from vertex
 - 3D trajectory reconstruction
- Particle identification
 - Muon track identify
 - CNN: flavor identification
- Zenith angle reconstruction: critical for L/E and NSI sensitivity



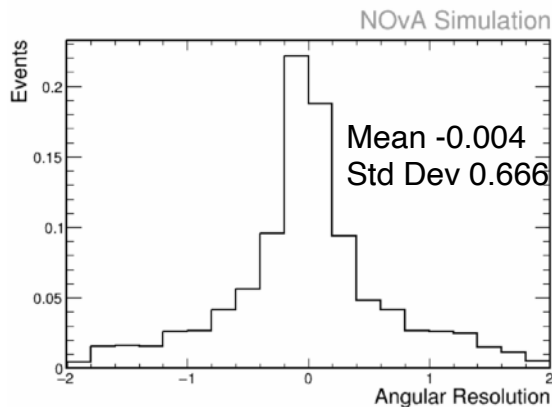
Angle Reconstruction

- Energy weighted vector sum of reconstructed prongs: $\vec{P}_{\text{tot}} = \frac{\sum_i E_i \vec{d}_i}{\left| \sum_i E_i \vec{d}_i \right|}$
- Zenith angle: $\cos\theta = \vec{P}_{\text{tot}} \cdot \hat{y}$ (vertical direction)
- Poor ν_μ angular residual

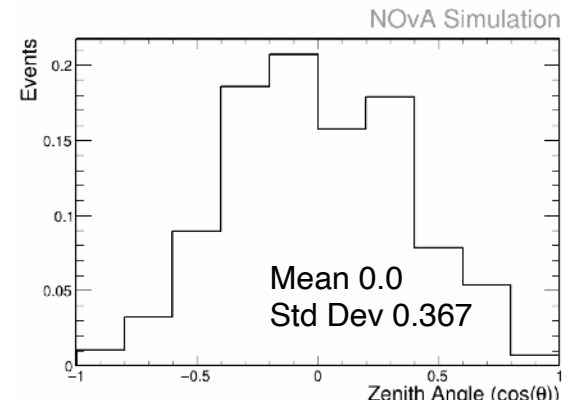
$\nu_\mu \theta_z$ distribution



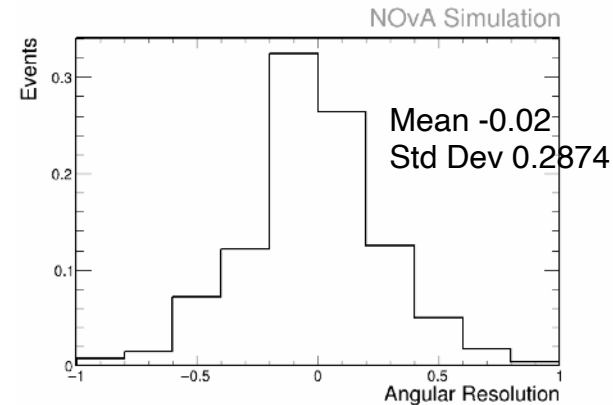
ν_μ residual



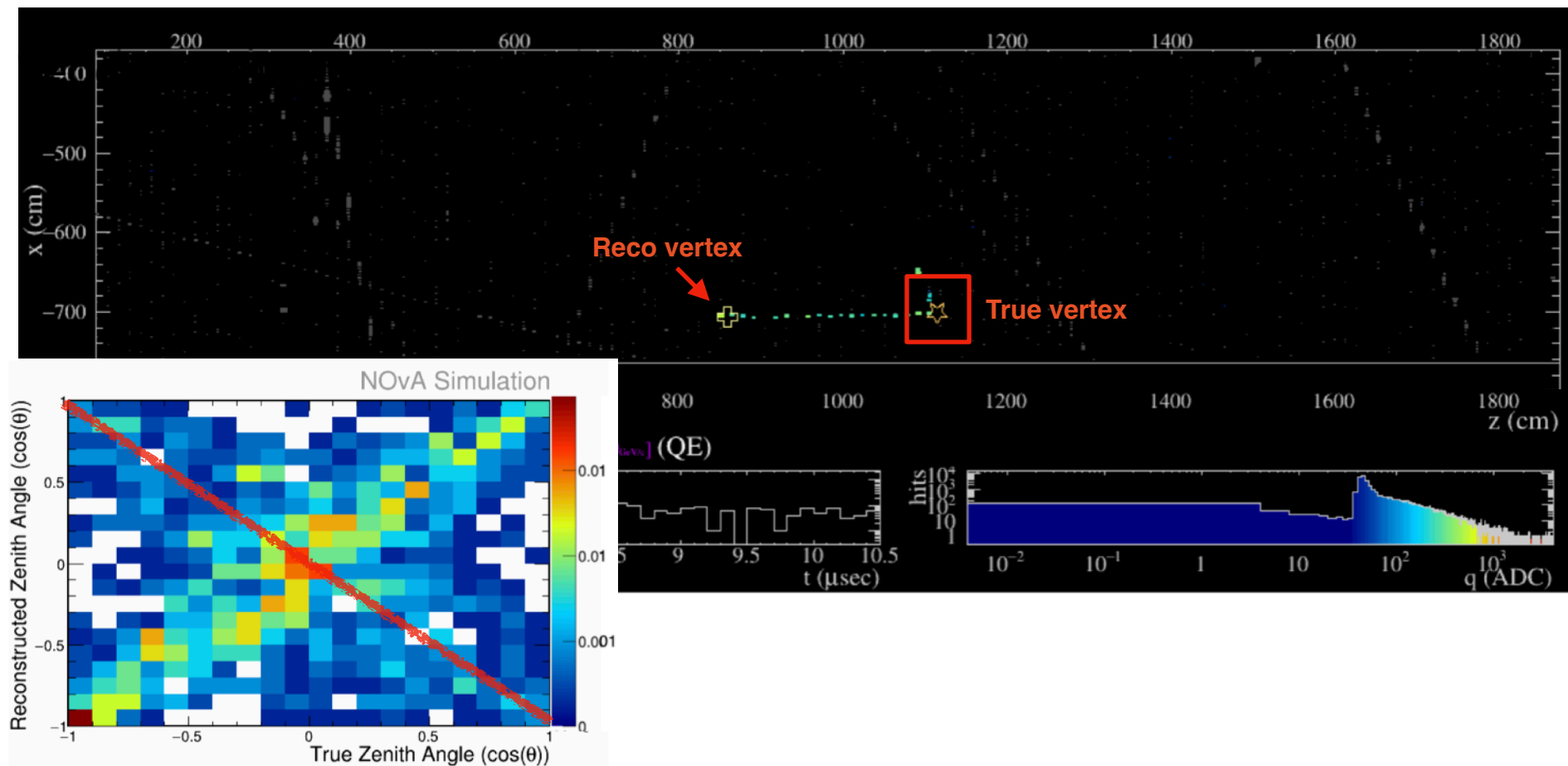
ν_e distribution



ν_e residual

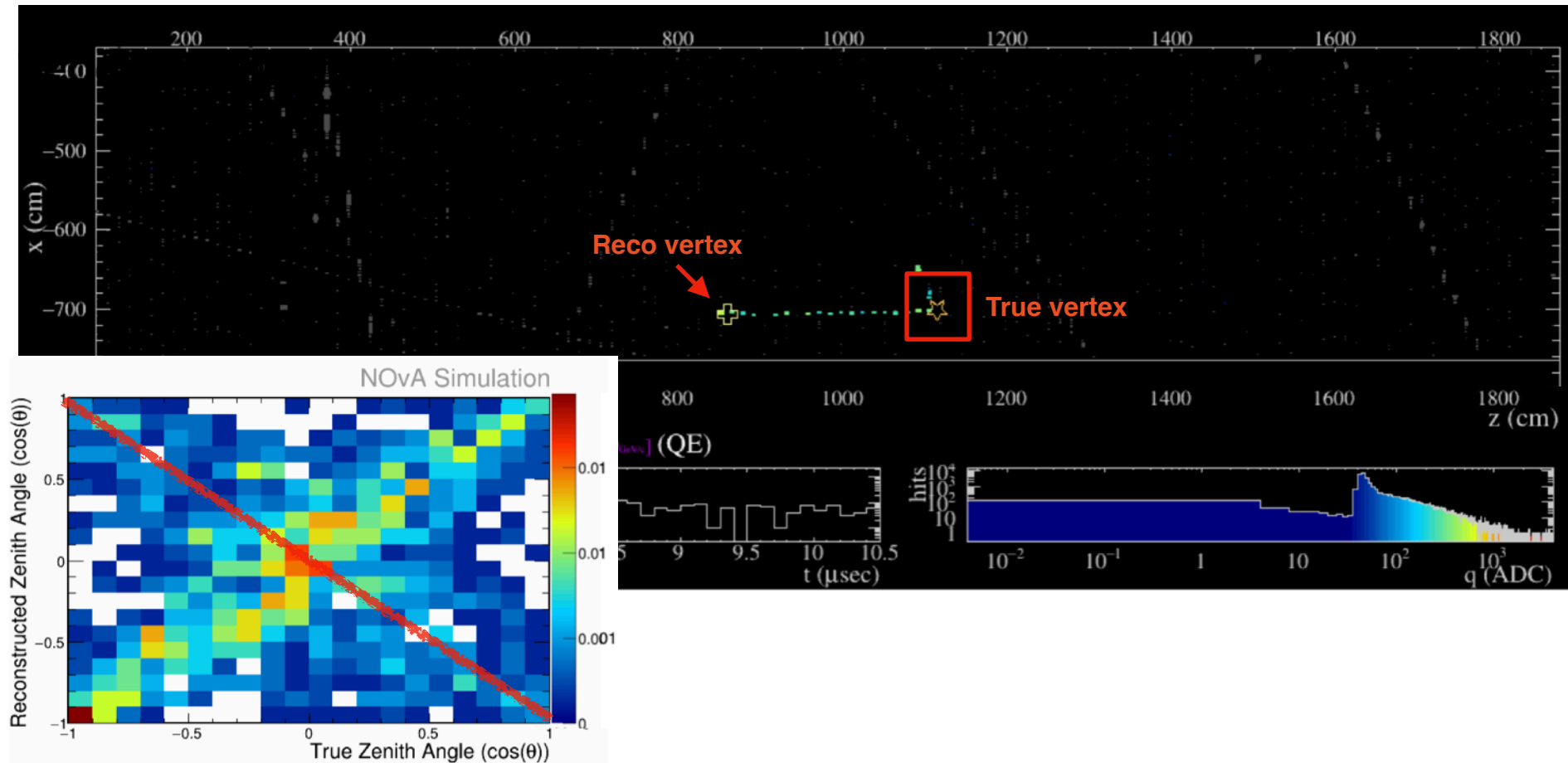


Reconstructed neutrino direction vs true neutrino direction



Reconstructed neutrino direction vs true neutrino direction

- Vertex reconstruction is biased toward upstream of beam direction



Optimistic ν_μ Angle Resolution

- Assuming we can overcome beam direction bias, what would resolution be?
- I study this by selecting well reconstructed events, where the reconstructed vertex is within 30cm of the true vertex
- Ways to achieve this in data: train vertex algorithm on simulated atmospheric neutrinos; look for hadronic activity to identify real vertex

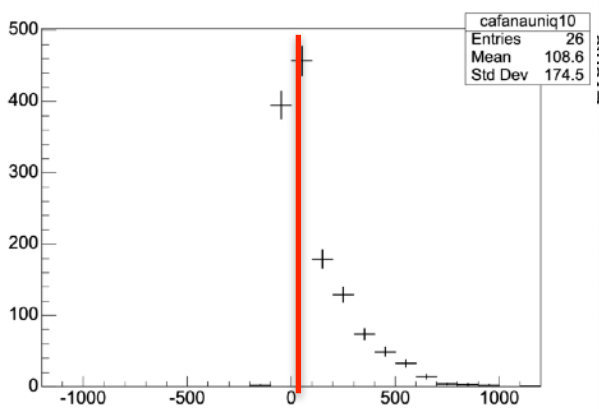
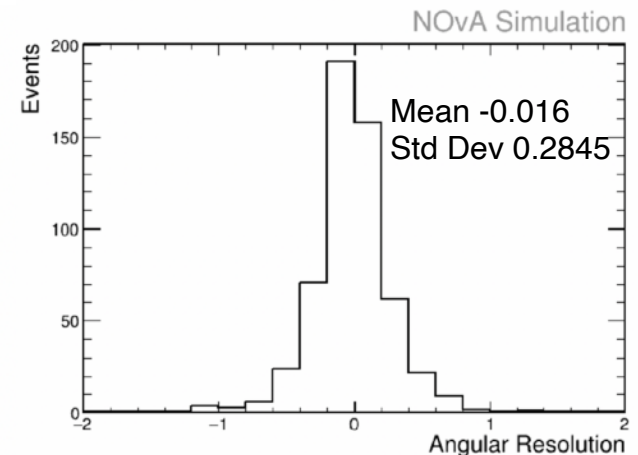
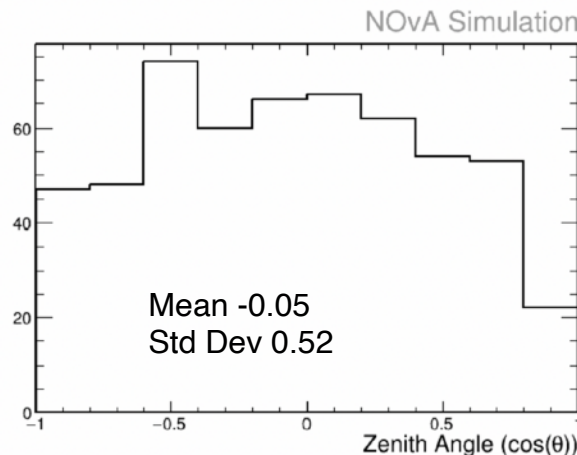


Figure 5.15: true neutrino vertex z position - reco neutrino vertex z position



ν_μ angular distribution

ν_μ angular residual

Can NOvA Far Detector see atmospheric neutrinos through the cosmic noise?

Selection Efficiency Summary

- Efficiency
$$\frac{\nu_{(reco\ with\ cut)}}{\nu_{(pass\ NNBar)}}$$
- Out of all neutrinos triggered, what fraction made it through the full analysis chain

Parameter	ν_e CC	ν_μ CC
Events after NNBar trigger	2,748	2,642
Events selected	200	862
Selection efficiency	7.3%	32.6%
NC contamination	1.9%	3.7%
Flavor cross-contamination	0.7%	4.0%

Table 5.2: Cut flow showing signal efficiency and background rejection for ν_μ CC selection

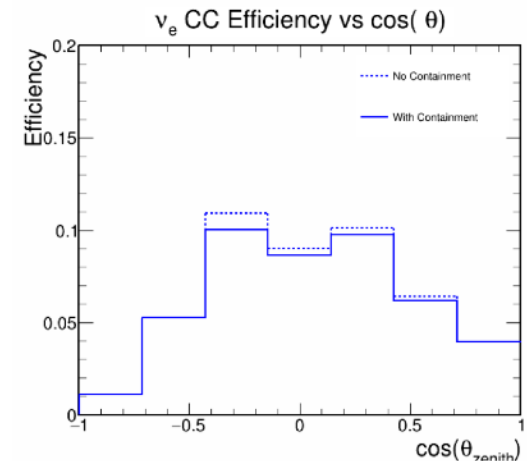
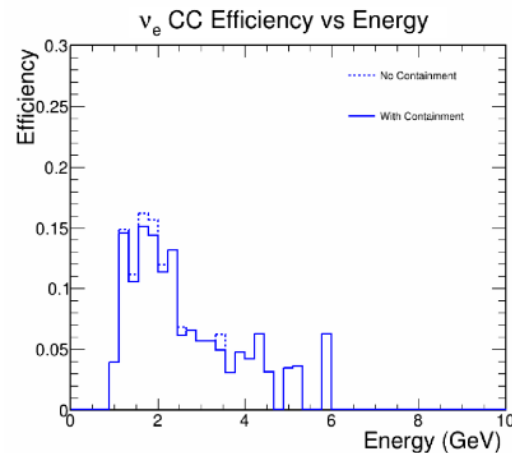
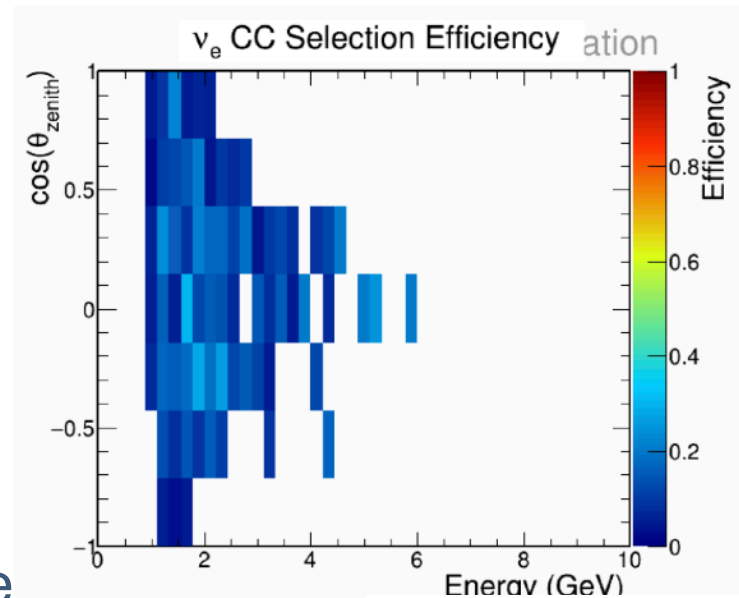
Cut Stage	Signal	Eff.	NC Bkg	Rej.	Cosmic Bkg	Rej.
After NNBar trigger	2662	—	1252	—	8.48×10^8	—
+ kHas1Prongs	2612	98.1%	1229	1.0×	8.48×10^8	1.0×
+ kNumuBasicQuality	2486	95.2%	1130	1.1×	7.79×10^8	1.1×
+ kNumuNCRej (RemID>0.75)	1042	41.9%	32	35.3×	7.79×10^8	1.0×
+ Containment	862	82.7%	32	1.0×	4.53×10^6	172.1×

Table 5.3: Cut flow showing signal efficiency and background rejection for ν_e CC selection

Cut Stage	Signal	Eff.	NC Bkg	Rej.	Cosmic Bkg	Rej.
After NNBar trigger	2763	—	1252	—	1.11×10^9	—
+ kHas1Prongs	2713	98.2%	1229	1.0×	1.11×10^9	1.0×
+ kNue2024FDBasicQuality	2386	87.9%	1013	1.2×	1.02×10^9	1.1×
+ kModifiedNue2024CorePart	1479	62.0%	627	1.6×	3.22×10^7	31.8×
+ kNue2024PID	209	14.1%	4	156.8×	6.35×10^3	5069.6×
+ Containment	200	95.70%	3	1.4×	6.35×10^3	1.0×

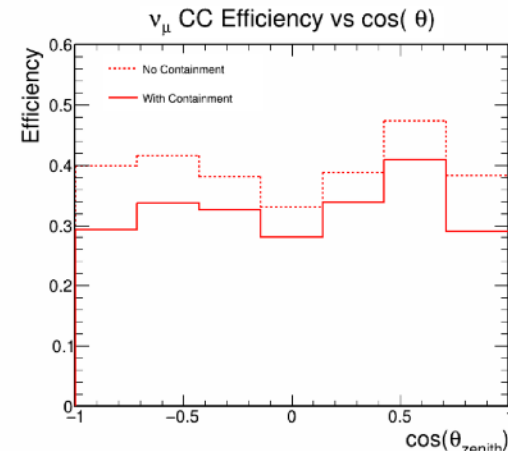
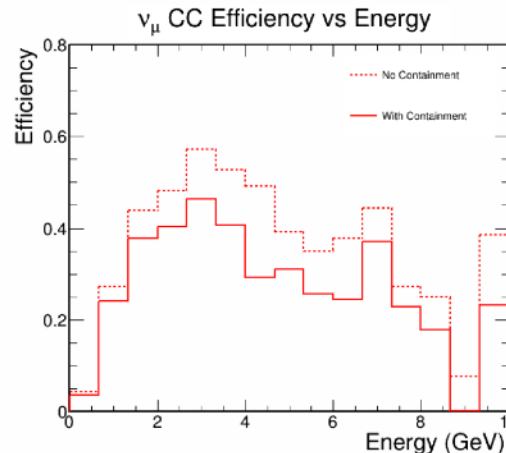
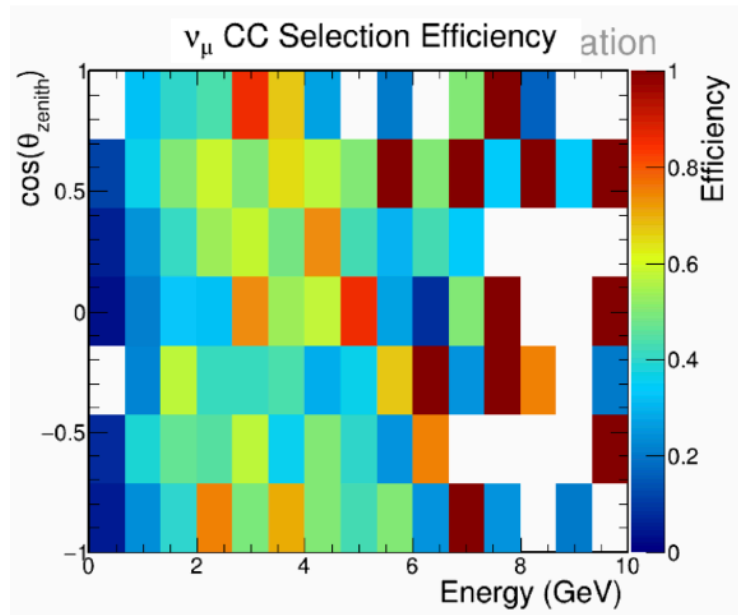
$\nu_e CC$ Selection Efficiency Continued

- 200 selected $\nu_e CC$
- 2748 total $\nu_e CC$
- Peak efficiency: $\sim 15\%$ at 1-2 GeV
- 7.3% total selection efficiency
- Strong zenith angle dependence



$\nu_\mu CC$ Selection Efficiency Continued

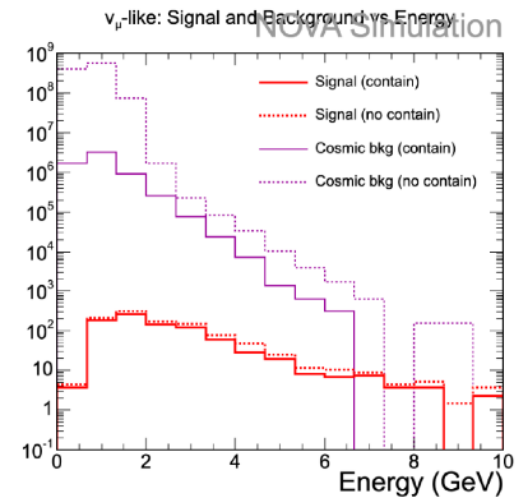
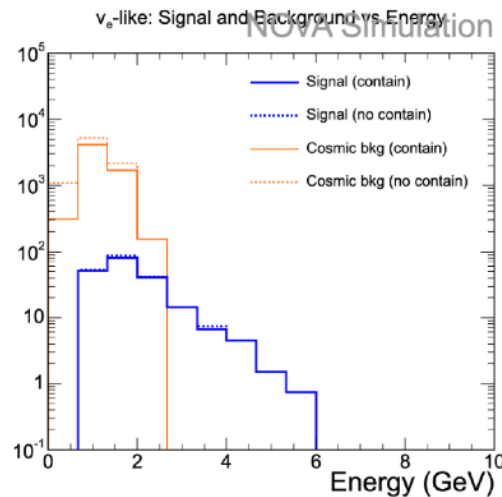
- 862 $\nu_\mu CC$ total
- $\nu_\mu CC$: 2642
- Stable 30-40% efficiency
1-6 GeV
- 32.6% total efficiency
- Weak zenith angle
dependence



Purity

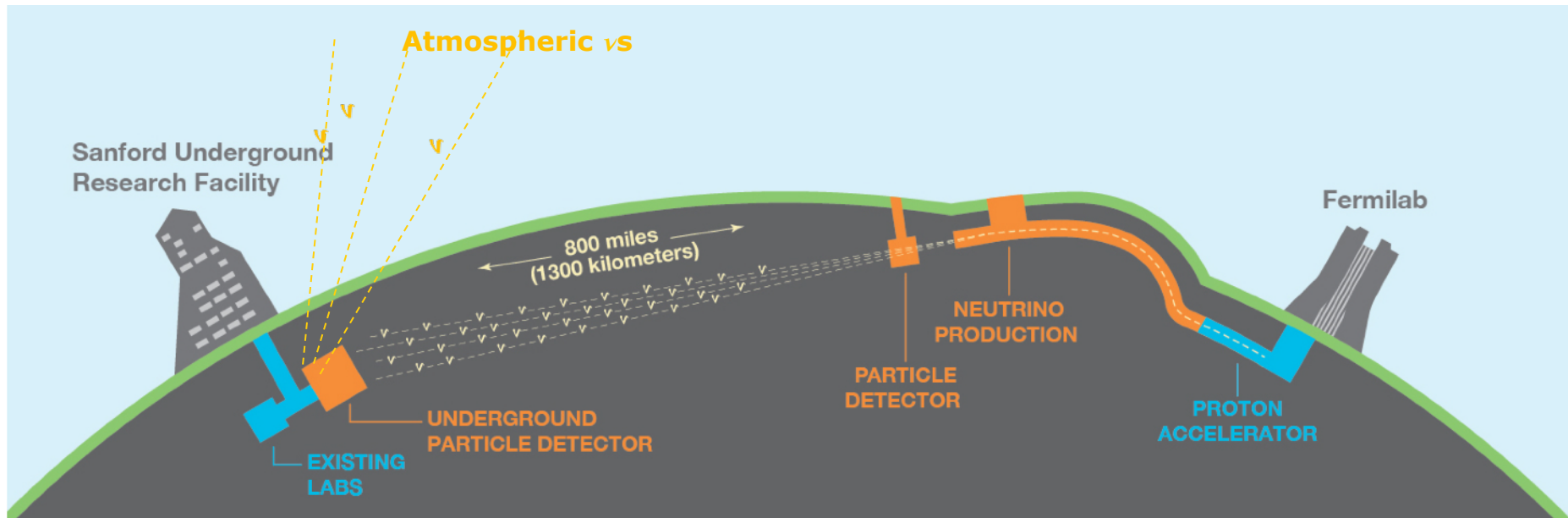
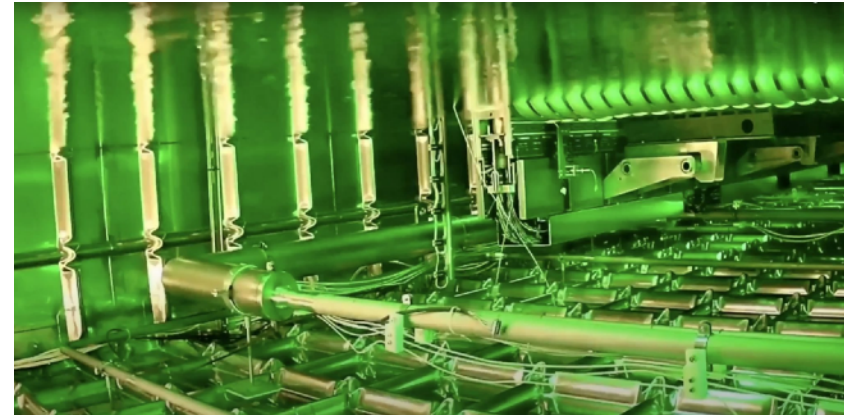
- Reality:
 - ν_e : 1 in 33 events is signal
 - ν_μ : 2 in 7000 events is signal
- Even with aggressive cuts, cosmic overwhelm ν_μ signals
- $E_{\nu_e} > 2.3$ GeV have good purity
- $E_{\nu_\mu} > 6.2$ GeV good purity, but hard to contain

Component	ν_e -like	ν_μ -like
<i>Signal Events (8 years)</i>		
Atmospheric ν signal	200	862
<i>Cosmic Ray Background</i>		
No containment cut	9×10^3	1×10^9
With containment cut	6×10^3	6×10^6
Rejection factor	$1.4 \times$	$172 \times$
<i>Purity</i>		
Purity (no containment)	2.36%	0.0001%
Purity (with containment)	3.06%	0.0141%



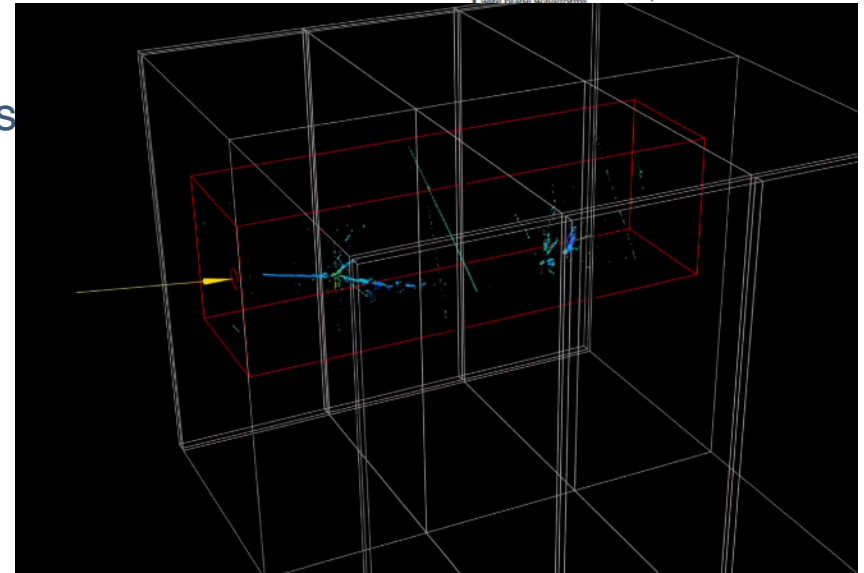
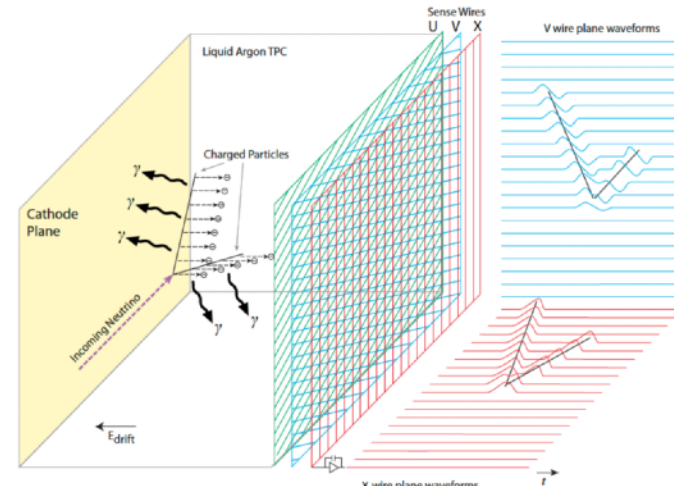
• Deep Underground Neutrino Experiment (DUNE)

- Key specs: 40 kton
- ~ 1.5 km underground for proton decay and supernova sensitivity
- liquid argon TPC
- Objective:
 - Extract ν oscillation parameters: mass ordering, δ_{CP}
 - Search for BSM physics
- Why DUNE for atmospheric neutrino:
 - Underground $\rightarrow \sim 10^6$ cosmic rejection (rock overburden)
 - Large mass: 40 kton vs 14 kton (NOvA)



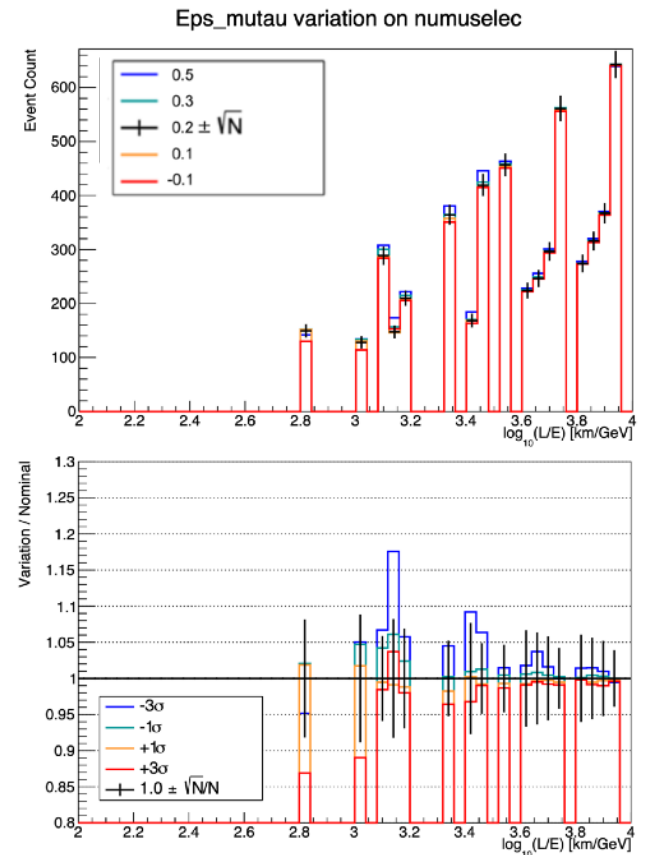
LArTPC Detector Design

- Incoming neutrino interact with argon nuclei, produce charged particles, scintillation light and ionization electrons
- Photon detection system determines interaction time t_0 to measure drift coordinates
- Ionization electrons drift in LAr liquid toward anode plane under uniform electric field.
- Use induced signal on sense wire planes to create two 2D track views
- Drift position, timing and charge info allow reconstruction of high resolution 3D particle trajectory
- Track length and calorimetry allows particle identification and energy estimation



Why MCMC

- Goal: Infer oscillation and NSI parameters from ν_{atm} event distribution. For each choice of parameters, predict what detector observe, and compare prediction with (Asimov) observed data
- Challenge: High-dimensional parameter space (θ_{SI} , θ_{NSI} , $\theta_{uncertainty}$)
- Solution: Instead of scanning every possible $\vec{\theta}$ combination on fixed grid, use Markov Chain Monte Carlo (MCMC) to sample posterior probability distribution efficiently



Markov Chain Monte Carlo

- Parameter space exploration method, it accepts and rejects proposed state based on ratio of likelihood or posterior probability value in the parameter space.

- At each Markov step:

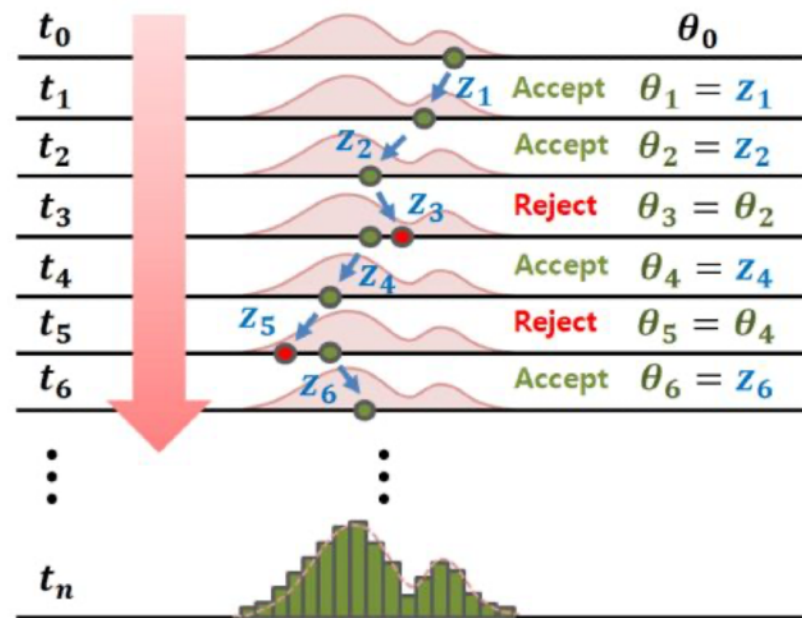
- Propose $\vec{\theta}$
- Compute oscillation probability $P_{\alpha \rightarrow \beta}(\vec{\theta})$
- Reweigh MC events
- Compare prediction to data
- Accept / reject proposed point

- Metropolis Hasting (a type of MCMC)

- Compute the acceptance ratio: $r = \frac{P(data|\theta') * P(\theta')}{P(data|\theta) * P(\theta)}$,

$$r(\theta_{new}, \theta_{t-1}) = \frac{Posterior(\theta_{new})}{Posterior(\theta_{t-1})}$$

- $p(\theta|D) \propto \mathcal{L}(D|\theta)\pi(\theta)$, $\mathcal{L}(D|\theta)$ calculates how well model predicts data



Bayesian MCMC Fit Strategy

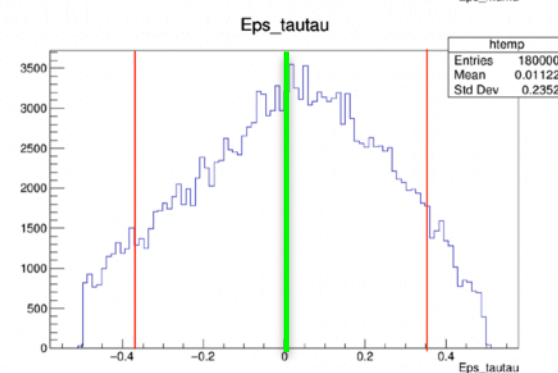
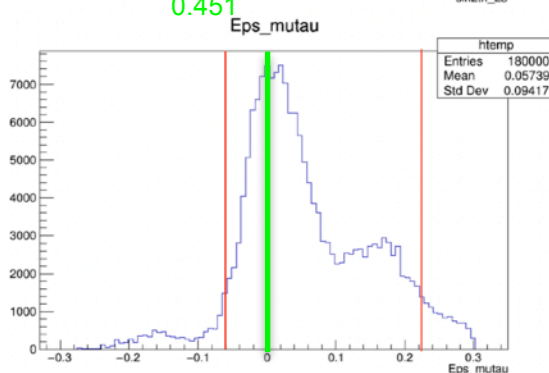
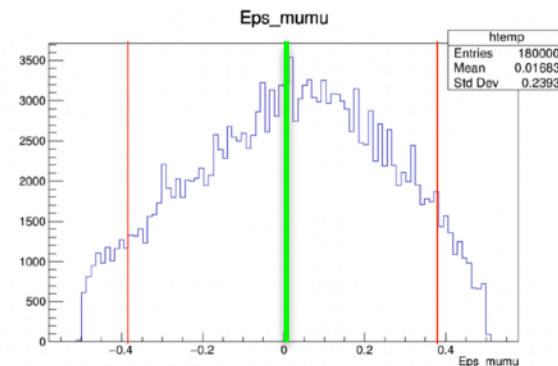
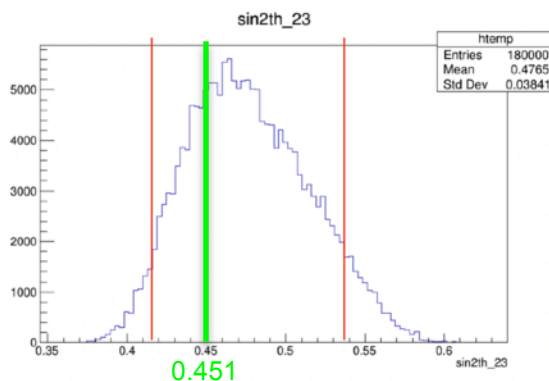
- Null hypothesis test: NSI sensitivity ($\text{NSI} = 0$)
 - Generate Asimov data with standard oscillation
 - Question: how far from 0 can we exclude ε_{NSI} ?
- Bayesian fit (5 Free parameters)
 - NSI ($\mu - \tau$ sector): $\varepsilon_{\mu\mu}, \varepsilon_{\mu\tau}, \varepsilon_{\tau\tau}$
 - atmospheric mixing: $\sin^2\theta_{23}$
 - Flux systematics: maximum error envelope spline-based nuisance parameters
- Fixed parameters:
 - All other oscillation parameters ($\Delta m_{ij}^2, \theta_{12}, \theta_{13}$, etc.)
- Output: 90% credible interval (Bayesian)

DUNE NSI sensitivity - Main Results

90% credible limits for NSI parameters (10 years, 400 kt·yr).

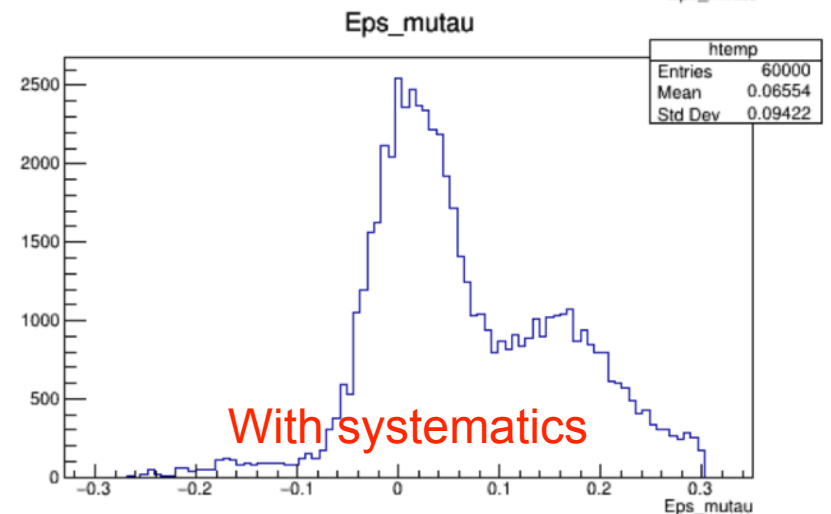
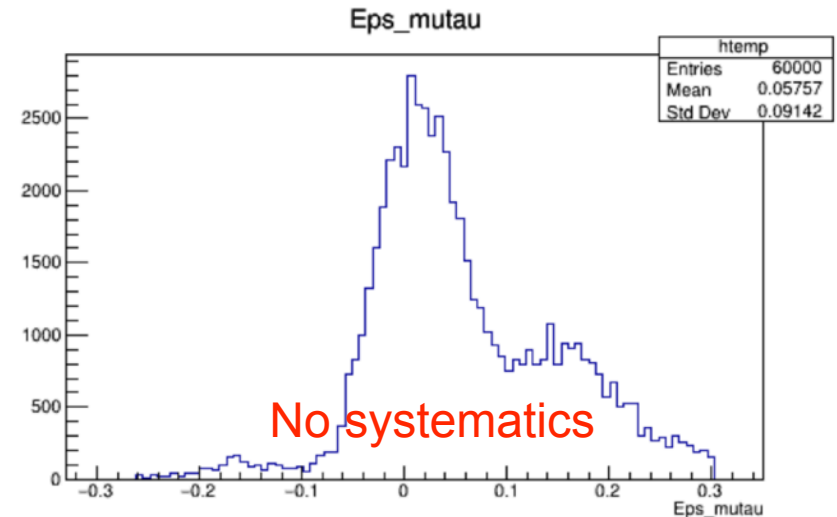
- 90% CI using 10 years of simulated atmospheric MC
- $|\varepsilon_{\mu\tau}| < 0.2$ consistent with Super-K constraint
- $\sin^2\theta_{23}$ at 0.477 ± 0.038 , consistent with global constraints

Parameter	Constraint
$ \varepsilon_{\mu\tau} $	< 0.20
$ \varepsilon_{\mu\mu} $	< 0.39
$ \varepsilon_{\tau\tau} $	< 0.39



Effect of Flux Systematics

- No systematics: $|\varepsilon_{\mu\tau}| < 0.198$
- With flux systematics: $|\varepsilon_{\mu\tau}| < 0.2$
- Flux systematics have negligible 1% change
- Flux uncertainties are negligible for this exposure
- Suggests statistics may be the dominant limitation, pending full systematic study

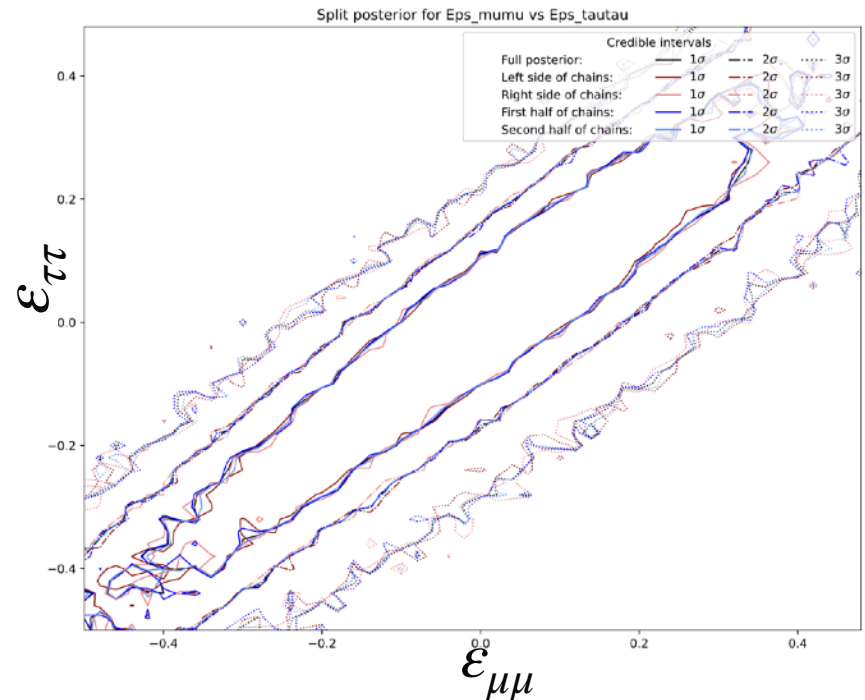
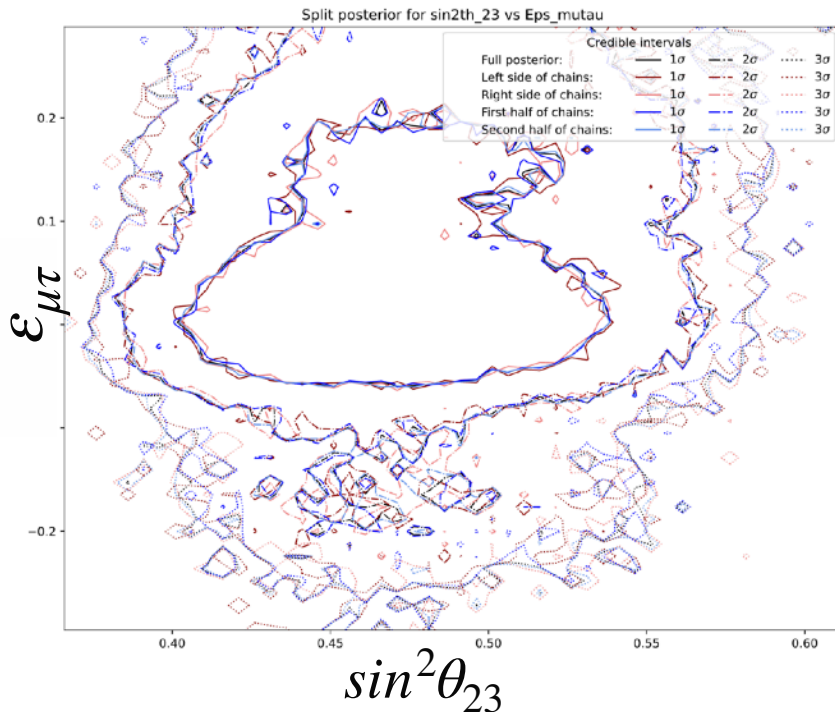
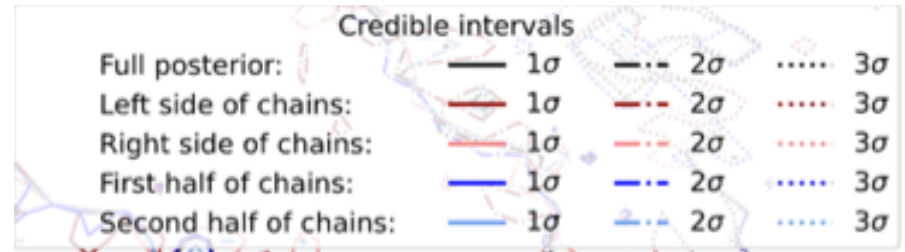


Parameter Correlations and Degeneracies

- Correlation coefficients:

- $r(\sin^2\theta_{23}, \varepsilon_{\mu\tau}) \sim 0.15$

- $r(\varepsilon_{\mu\mu}, \varepsilon_{\tau\tau}) \sim 0.94$



Future Perspective

- NOvA
 - ν_e sample: has better potential to be sensitive to NSI, but improved cosmic rejection below 3GeV should be explored
 - ν_μ sample: need better cosmic rejection below 7GeV with improved vertex reconstruction
- DUNE
 - More exposure
 - Use higher energy (above 10 GeV)
 - Combine DUNE beam + atmospheric
 - Reduce $\sin^2\theta_{23}$, $\varepsilon_{\mu\tau}$ degeneracy
 - Reduce $\varepsilon_{\mu\mu}$, $\varepsilon_{\tau\tau}$ degeneracy
 - Joint analysis leverage complementary systematics
 - When DUNE takes data:
 - Validate NSI sensitivity with real data

Summary & Conclusion

- NOvA atmospheric analysis
 - Developed the first atmospheric neutrino analysis in NOvA
 - Surface operation makes cosmic rejection the dominant challenge
 - Achieved selected-sample purities of:
 - ν_μ CC: 0.014%
 - ν_e CC: 3.06%
- DUNE atmospheric NSI sensitivity
 - Atmospheric neutrinos provide sensitivity to matter-driven NSI effects
 - Projected sensitivity reaches $|\varepsilon_{\mu\tau}| < 0.2$ (consistent with current limits)
 - Strong $\varepsilon_{\mu\mu}, \varepsilon_{\tau\tau}$ degeneracy weakens sensitivity to diagonal NSI
- Key contribution
 - Built NOvA atmospheric analysis chain from scratch
 - Developed atmospheric flux systematic framework for DUNE
 - Produced a full DUNE atmospheric NSI sensitivity study

Thank you!



Backup

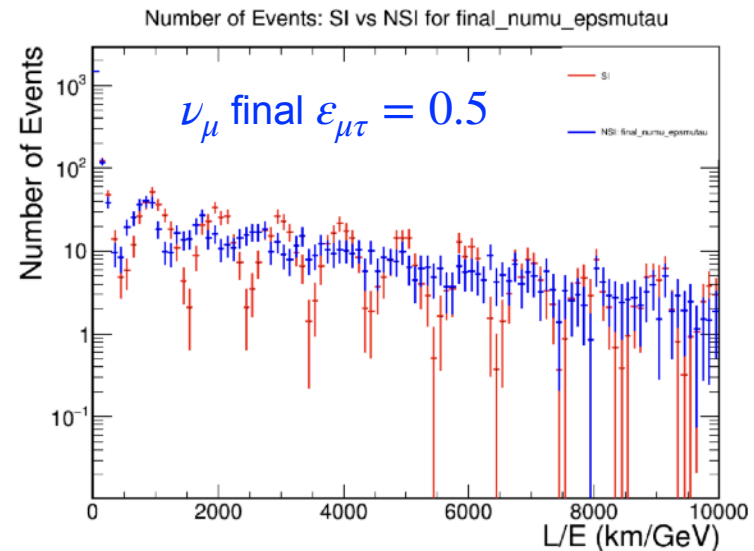
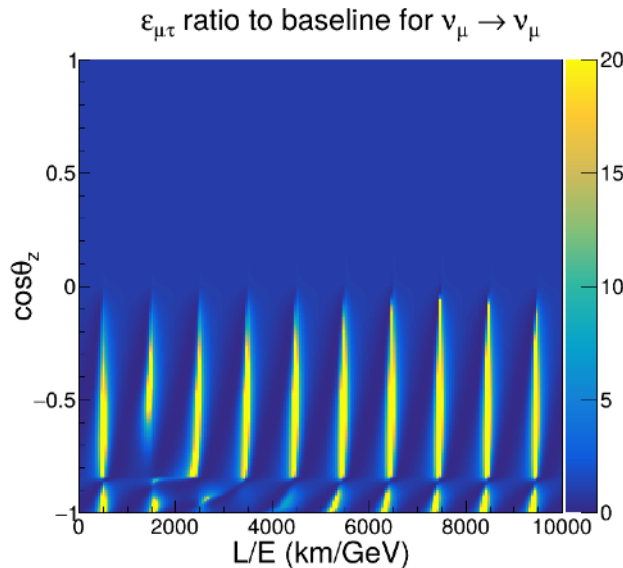
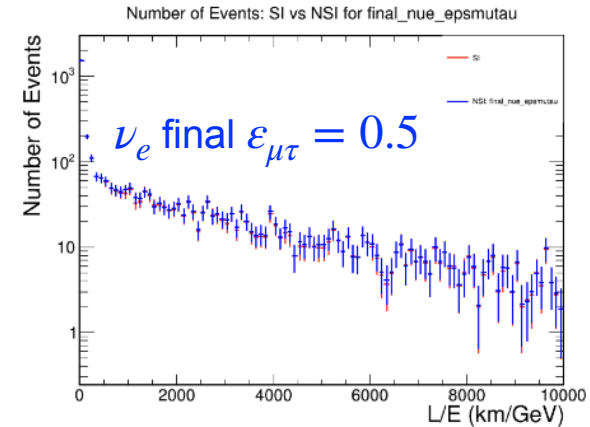


Comparison to Current Limits

- My results: (with e^- coupling)
 - $\varepsilon_{\mu\tau}$: $[-0.060, 0.229]$
 - $\varepsilon_{\mu\mu}$: $[-0.394, 0.393]$
 - $\varepsilon_{\tau\tau}$: $[-0.396, 0.382]$
 - $\varepsilon_{\tau\tau} - \varepsilon_{\mu\mu}$: $[-0.138, +0.119]$
 - Exposure: 10 years, 0.1 to 10 GeV
- Current Global NSI limits:
 - $\varepsilon_{\mu\tau}$: $[-0.0063, +0.016]$ with e^- coupling
 - $\varepsilon_{\mu\mu}$: $[-0.2, +0.1]$
 - $\varepsilon_{\tau\tau}$: $[-0.17, +0.093]$
 - $\varepsilon_{\tau\tau} - \varepsilon_{\mu\mu}$: $[-0.040, +0.047]$
 - arXiv:1805.04530 2020
- NOvA beam
 - $\varepsilon_{\mu\tau}$ not included in study
- IceCube (universal coupling)
 - $\varepsilon_{\mu\tau}$: $[-0.0041, +0.0031]$
 - 305,735 up-going ν_μ 500 GeV to 9976 GeV
- Super-K atmospheric
 - $\varepsilon_{\mu\tau}$: $[-0.046, +0.019]$
 - $\varepsilon_{\tau\tau} - \varepsilon_{\mu\mu}$: $[-0.050, +0.138]$
 - 15 years exposure
 - Mahdi Taani. “Non-Standard Neutrino Interaction Analysis with Atmospheric Neutrino Data in Super-Kamiokande I-IV and the Design of the Hyper-Kamiokande Outer Detector”

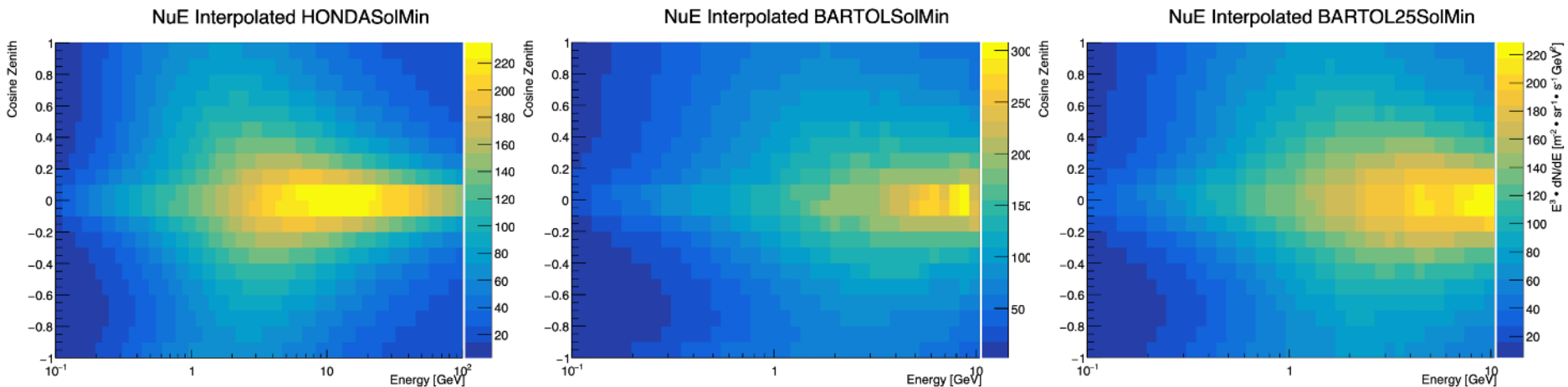
NSI Signals in 1D Projection

- Event rate match Oscillogram pattern
- NSI show significant deviation from SI after statistical error



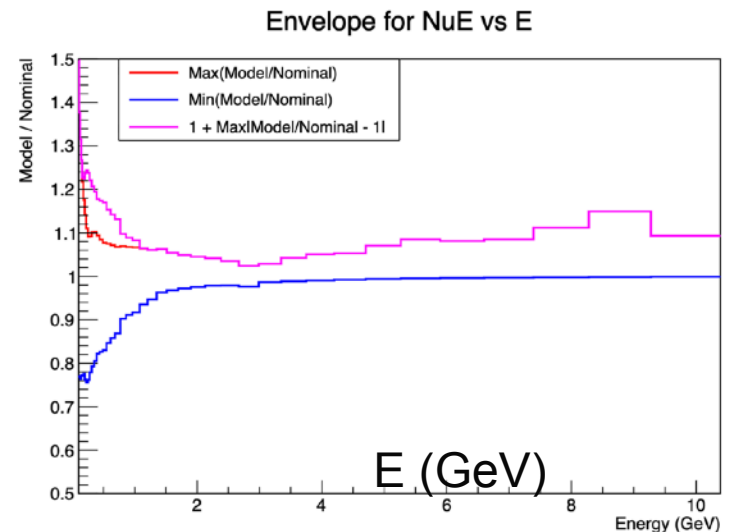
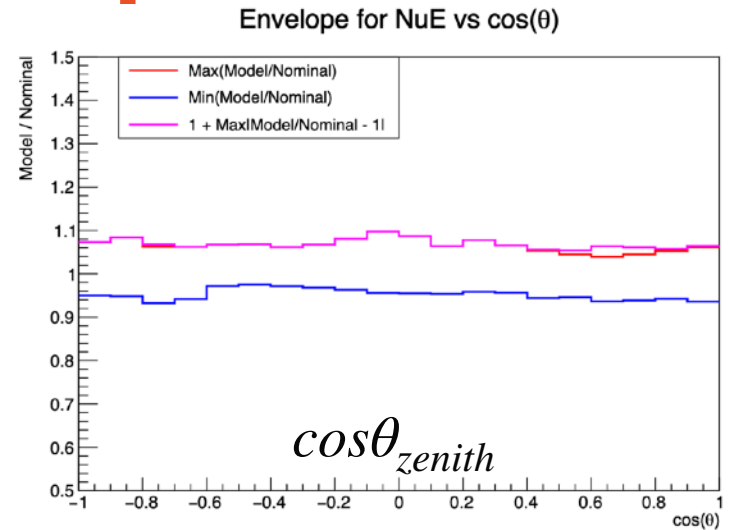
Flux Systematics

- Goal: Build a realistic NSI sensitivity study for DUNE atmospheric neutrinos
- Challenge: Atmospheric group had no systematics implemented
- Implemented model to model flux uncertainties (Honda vs Bartol)
- Largest uncertainty for atmospheric samples
- Energy & zenith bin normalization



Flux ratio model comparison

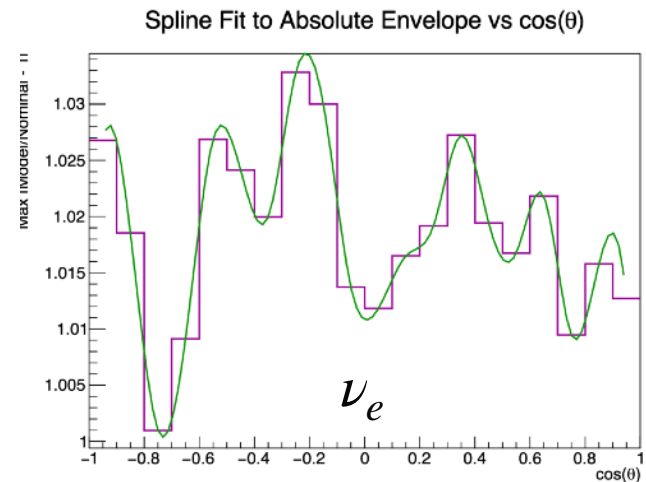
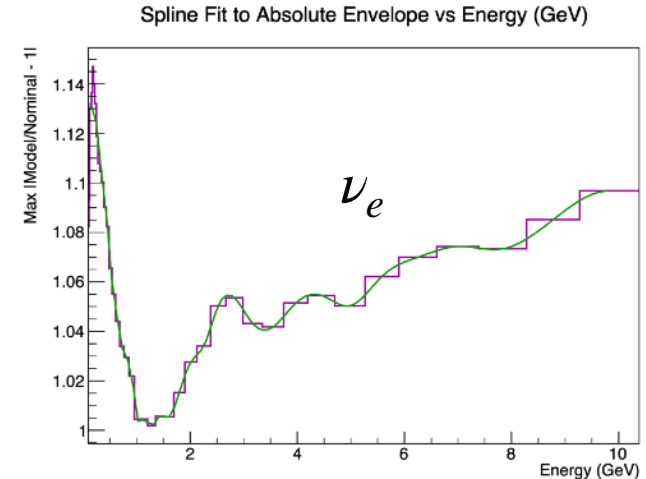
- 3 flux models compared:
 - Honda, Bartol, Bartol 2025
- Model differences:
 - Up to 20-30% in energy/zenith regions
- Maximum error envelope approach:
 - Take maximum deviation across all models
 - Conservative systematic treatment



• HONDASolMin as nominal

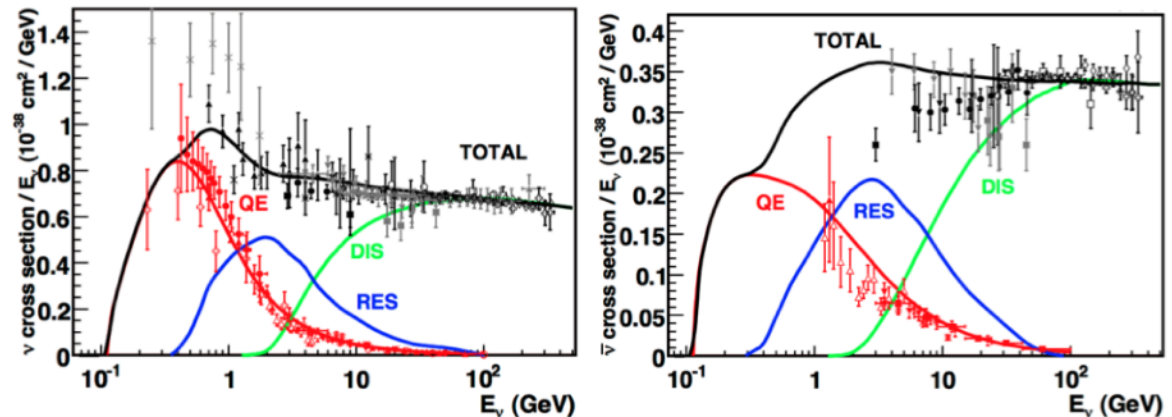
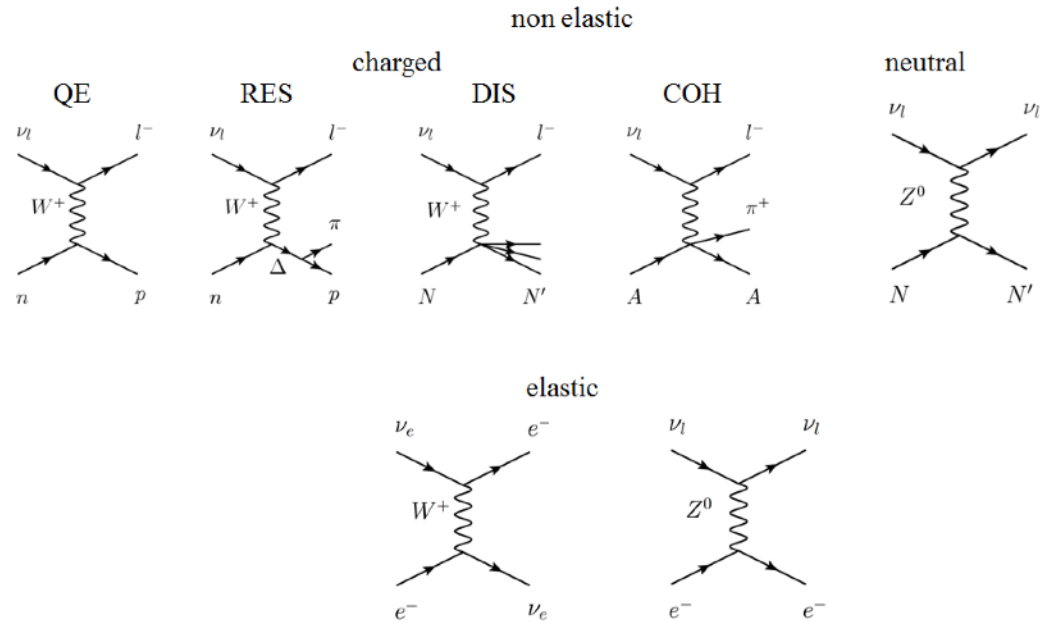
Spline Parameterization

- Need smooth flux functions for oscillation fits
 - TSpline3 interpolation through envelope
- Implementation in MaCH3:
 - Flux systematics become nuisance parameters
 - Spline nodes constrained by Gaussian prior (mean = 1, $\sigma = 0.1$)
 - Fit deforms flux smoothly within allowed uncertainties



Interaction Models

- CCQE
 - Dominant at $\sim 1\text{GeV}$
- Resonance Pion Production (RES)
- Deep Inelastic Scattering (DIS)



Signal and Background Estimation

- Signal: $N_{events} = \Phi_{\nu} \times \sigma_{\nu N} \times N_{targets} \times \epsilon_{trigger} \times T_{exposure}(8yr)$
 - $\nu_e CC$: 2748
 - $\nu_{\mu} CC$: 2642
- Background:
 - NC: 1245
 - NNBarDDT run at 8 Hz
 - Standard approach: run NNBar on cosmic data and then reconstruction
 - Technical issue: software upgrade killed NNBar trigger simulation
 - Solution: emulated NNBar selection with software cut
 - 6,000 ν_e like cosmic background
 - 6×10^6 ν_{μ} like background

NSI Lagrangian

$$\mathcal{L}_{\text{NC-NSI}} = -\sqrt{2}G_F \sum_{f,P,\alpha,\beta} \epsilon_{\alpha\beta}^{fP} (\bar{\nu}_\alpha \gamma^\mu P_L \nu_\beta) (\bar{f} \gamma_\mu P f) \quad (2.32)$$

$$\mathcal{L}_{\text{CC-NSI}} = -\sqrt{2}G_F \sum_{f,P,\alpha,\beta} \epsilon_{\alpha\beta}^{fP} (\bar{\nu}_\alpha \gamma^\mu P_L \nu_\beta) (\bar{f} \gamma_\mu P f') \quad (2.33)$$

where G_F is the Fermi constant, f, f' are the first generation fermion (e, u, d) for SM. P is the chiral projector $L, R = (1 \pm \gamma^5)/2$, α and β are two flavors of neutrino. $\epsilon_{\alpha\beta}^{fP}$ represents the dimensionless NSI coupling strength relative to G_F for neutrinos interacting with fermion f with chirality P . $\epsilon_{\alpha\beta}^{fP} \propto \mathcal{O}(\frac{G_M}{G_F})$, G_M is the coupling constant.

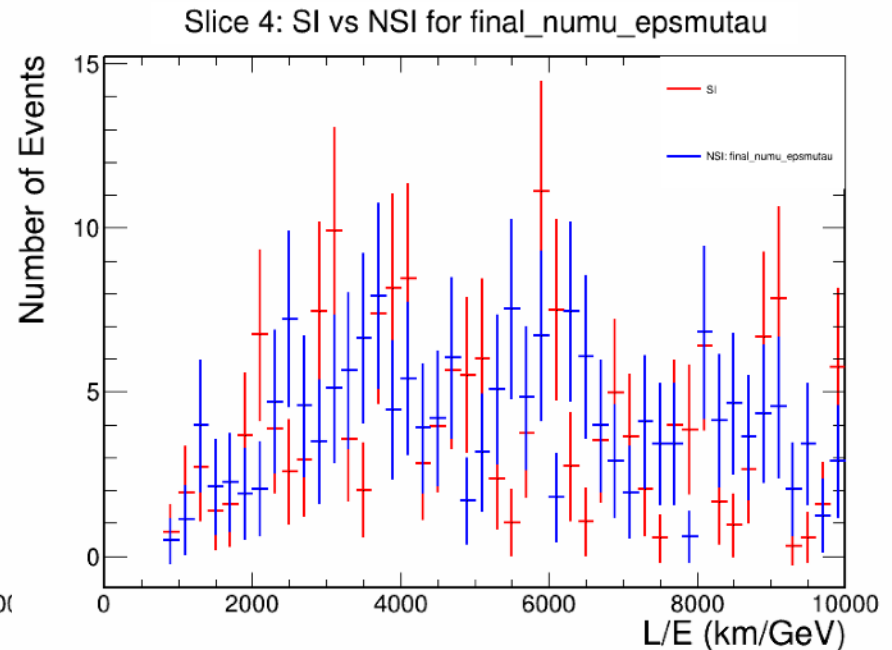
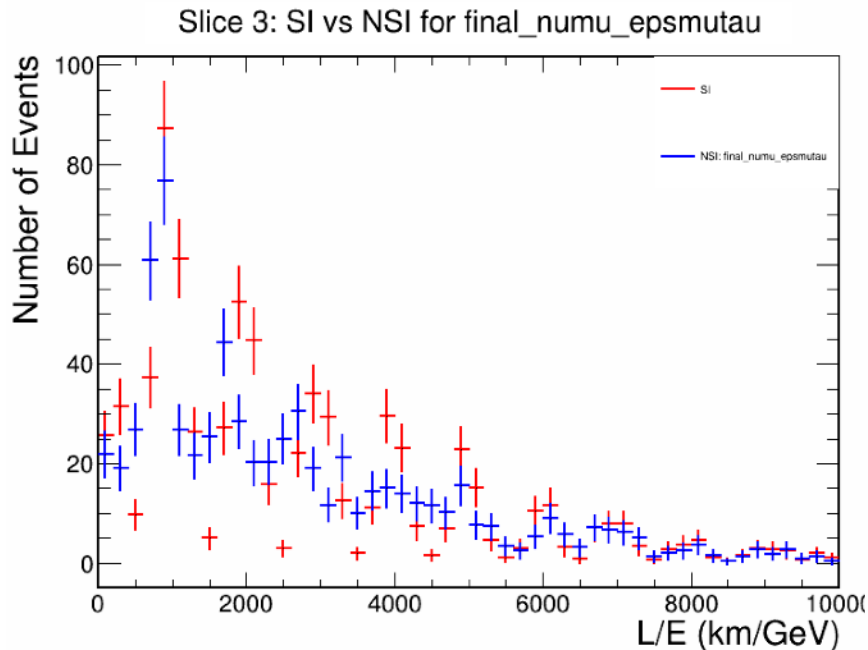
$$\epsilon_{\alpha\beta} = \sum_{f=e,u,d} \left(\epsilon_{\alpha\beta}^{fL} + \epsilon_{\alpha\beta}^{fR} \right) \frac{N_f}{N_e}$$

NSI Signals in 1D Slices

- NSI effect is angle dependent
- Slice $0 < \cos(\theta) < 1$ show no NSI impact
- Challenge: Can we extract this subtle signal from cosmic ray background?

Slice 3: $-0.5 < \cos(\theta) < 0$ (horizontal to upward)

Slice 4: $-1 < \cos(\theta) < -0.5$ (strongly upward) - maximum Earth matter traversal

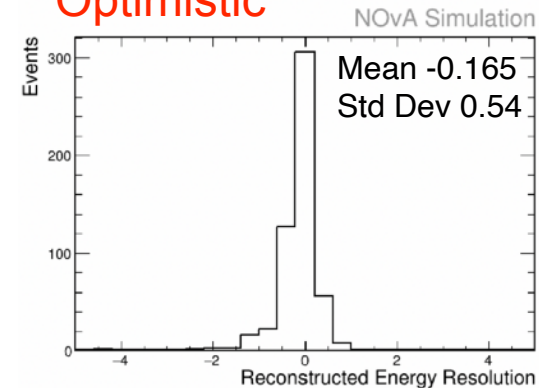
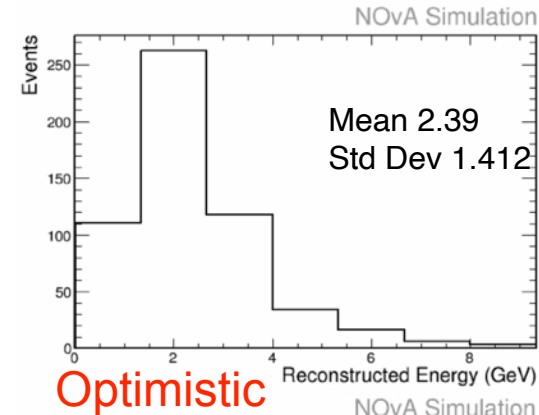
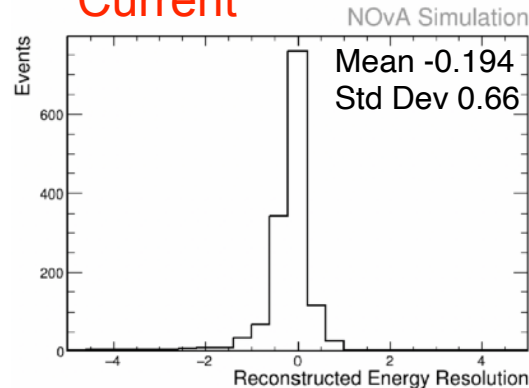
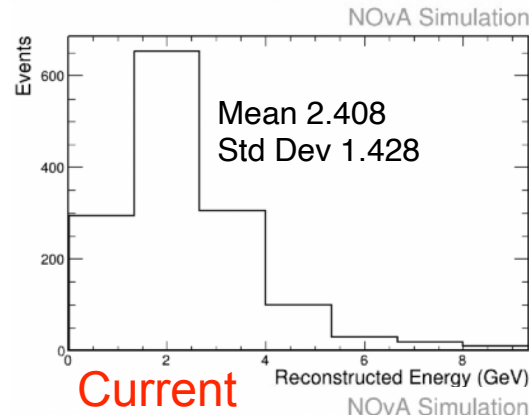


Detailed NOvA cosmic estimate

- Standard Method: run NNBar DDT trigger simulation on raw cosmic data, then apply reconstruction and analysis cuts to measure final cosmic background rate.
- Technical limitations: software incompatibility: artdaq-core v3.07.04 -> v3.09.02 brakes NNBarDDT
- Workaround: apply software cut (kNNBarCutCosmic) to estimate NNBar topology
- Measured rate: 218.2 Hz
- Normalization with NNBar DDT actual rate: 8 Hz
- Scale factor $8/218.2 = 0.0367$ -> 3.67% compared to 0.81% NNBarDDT on cosmic
- Higher passing rate due to reconstruction applies cut on neutrino and get rid of cosmic already

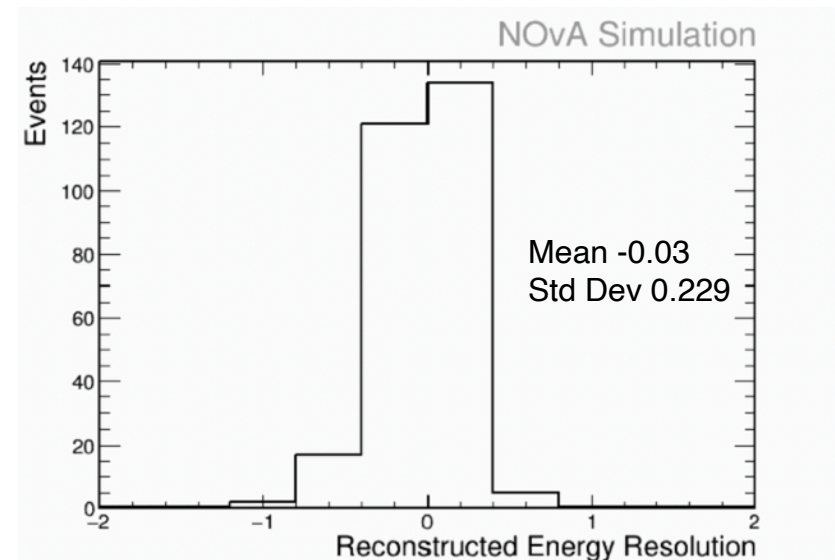
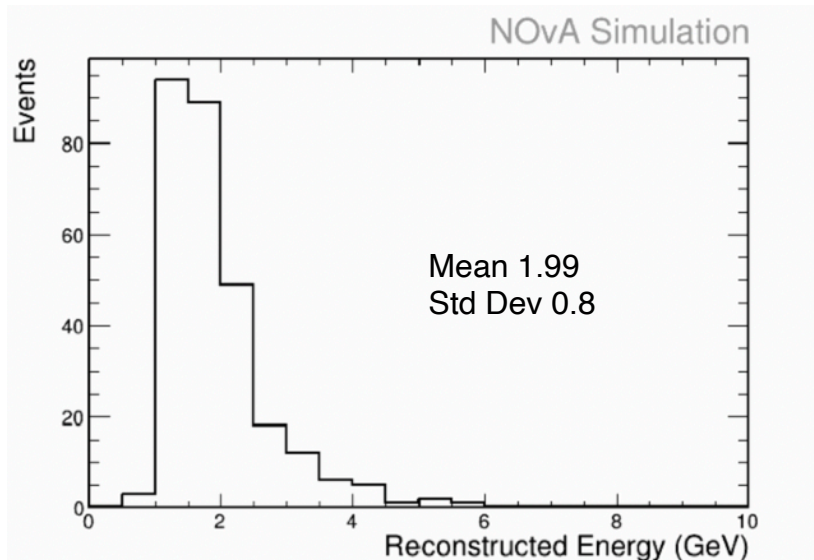
ν_μ Energy Reconstruction

- Why it matters: oscillation depends on L/E
- $E_{\nu_\mu} = E_\mu$ (from track length) + E_{had} (calorimetry)
- NOvA beam neutrino in similar energy range
- Challenge: muons can exit detector



ν_e Energy Reconstruction

- E_{ν_e} = calibrated sum of EM shower + E_{had} (calorimetry)
- σ_E/E at 2 GeV $\sim 11\%$ fractional energy resolution, compared to muon at 33% (current) and optimistic 27% (optimistic)



Analysis Performance Metrics

- Two key questions for any atmospheric neutrino analysis
 - How many signal events we can select
 - How clean is the selected sample
- Selection efficiency:
 - Efficiency = selected ν / all ν passing trigger
 - Why it matters: higher efficiency $\rightarrow$ more signal $\rightarrow$ better statistical power
 - Trade-off: loose cut = high efficiency but poor purity
- Signal purity:
 - Purity = signal / (signal + background)
 - Why it matters: higher purity $\rightarrow$ easier to extract oscillation signal
 - Trade-off: tighter cuts = low efficiency $\rightarrow$ lower statistics
- Optimization: maximize significance ($S/\sqrt{B}$)

Cut flow

- Systematic selection optimization for each flavor
- ν_μ selection (top)
 - Quality cut: maintain high efficiency (93%)
 - ReMId: 35x NC rejection at cost of efficiency (42%)
 - Containment: critical for cosmic 128x rejection
- ν_e selection (bottom)
 - Energy/topology core cuts: 32x cosmic rejection
 - PID: extremely powerful 5070x cosmic, 157x NC rejection
 - Only 7.3% efficiency but excellent bkg rejection

Table 5.2: Cut flow showing signal efficiency and background rejection for ν_μ CC selection

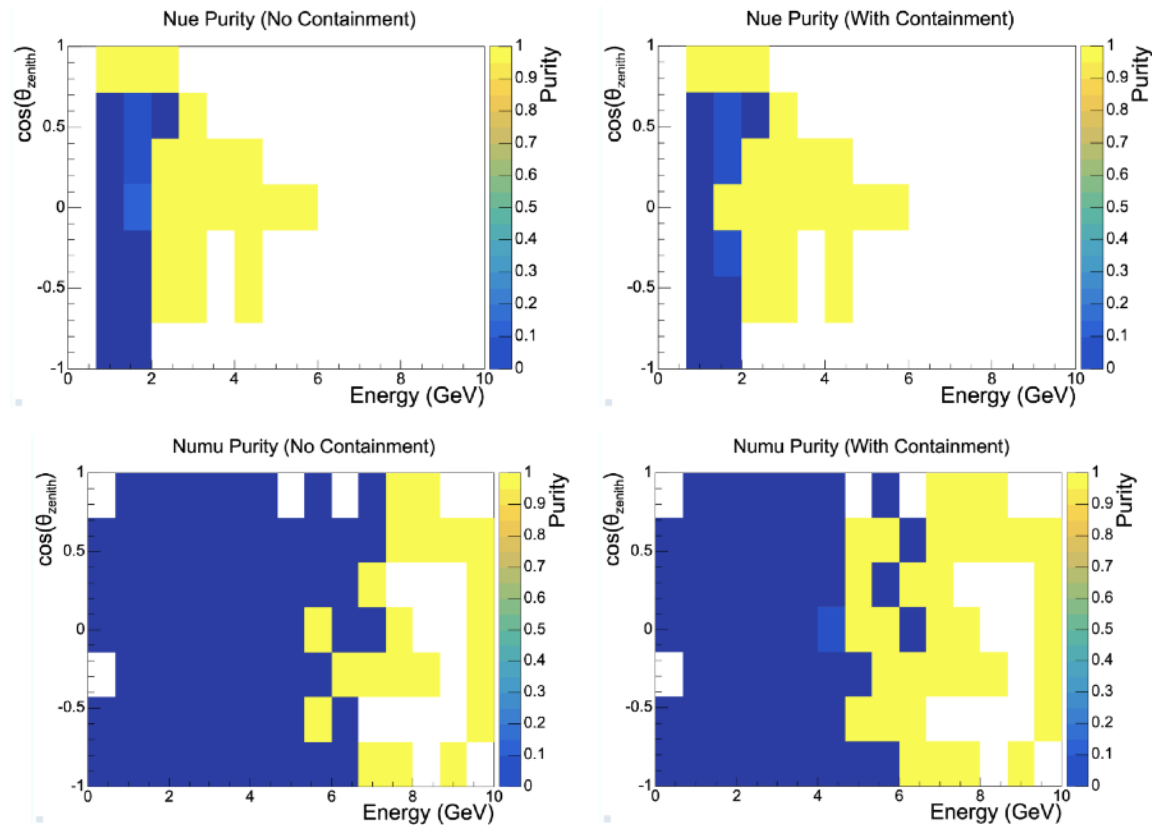
Cut Stage	Signal	Eff.	NC Bkg	Rej.	Cosmic Bkg	Rej.
After NNBar trigger	2662	—	1252	—	8.48×10^8	—
+ kHas1Prongs	2612	98.1%	1229	1.0×	8.48×10^8	1.0×
+ kNumuBasicQuality	2486	95.2%	1130	1.1×	7.79×10^8	1.1×
+ kNumuNCRej (RemID>0.75)	1042	41.9%	32	35.3×	7.79×10^8	1.0×
+ Containment	862	82.7%	32	1.0×	4.53×10^6	172.1×

Table 5.3: Cut flow showing signal efficiency and background rejection for ν_e CC selection

Cut Stage	Signal	Eff.	NC Bkg	Rej.	Cosmic Bkg	Rej.
After NNBar trigger	2763	—	1252	—	1.11×10^9	—
+ kHas1Prongs	2713	98.2%	1229	1.0×	1.11×10^9	1.0×
+ kNue2024FDBasicQuality	2386	87.9%	1013	1.2×	1.02×10^9	1.1×
+ kModifiedNue2024CorePart	1479	62.0%	627	1.6×	3.22×10^7	31.8×
+ kNue2024PID	209	14.1%	4	156.8×	6.35×10^3	5069.6×
+ Containment	200	95.70%	3	1.4×	6.35×10^3	1.0×

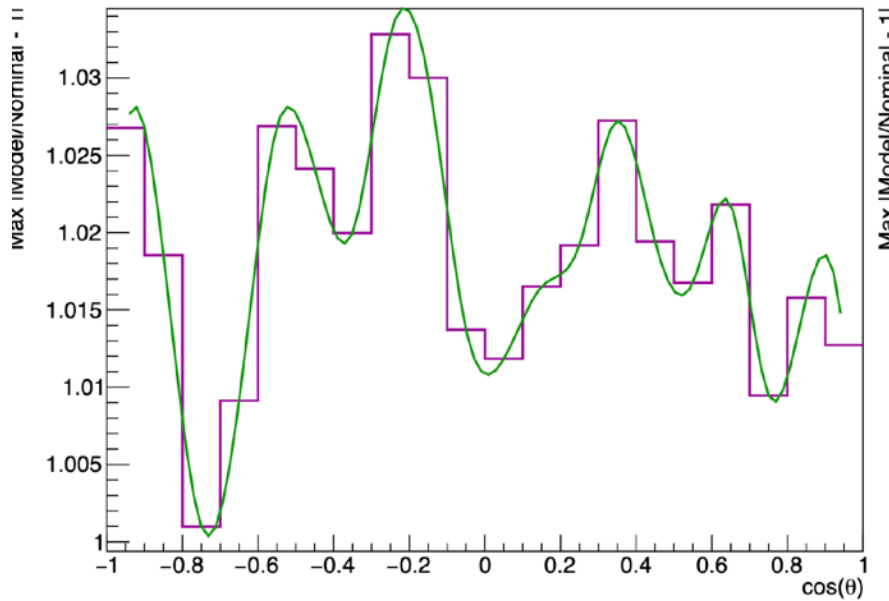
NOvA 2D Purity plots

- 2D purity shows overwhelming cosmic background

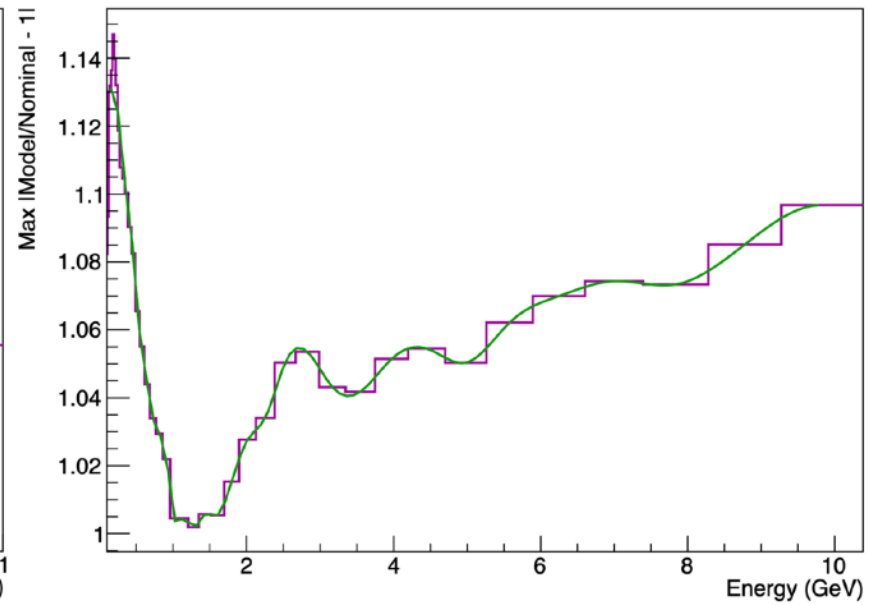


ANuM

Spline Fit to Absolute Envelope vs $\cos(\theta)$

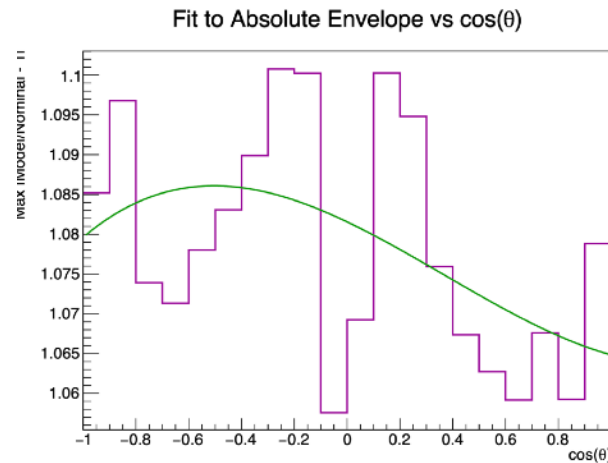
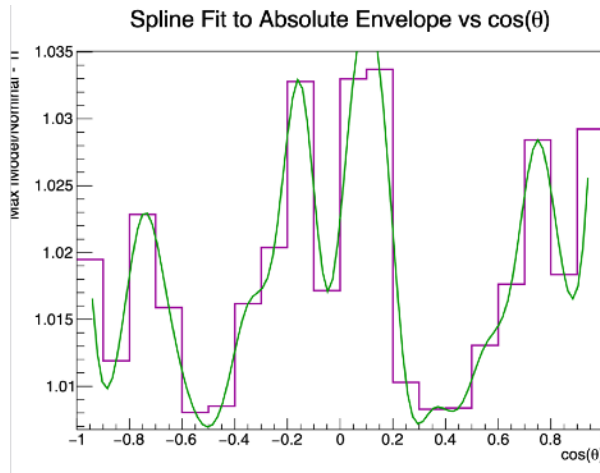


Spline Fit to Absolute Envelope vs Energy (GeV)

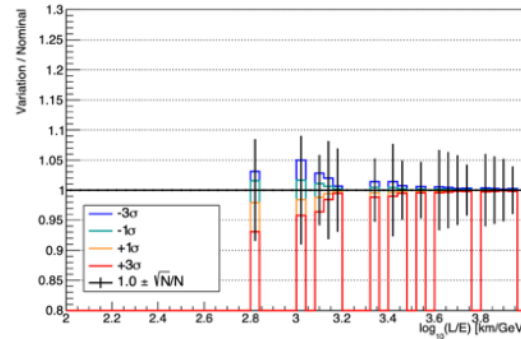
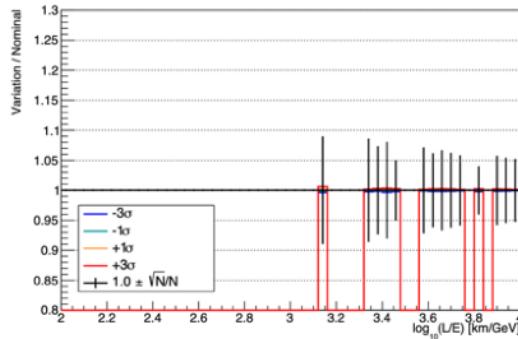
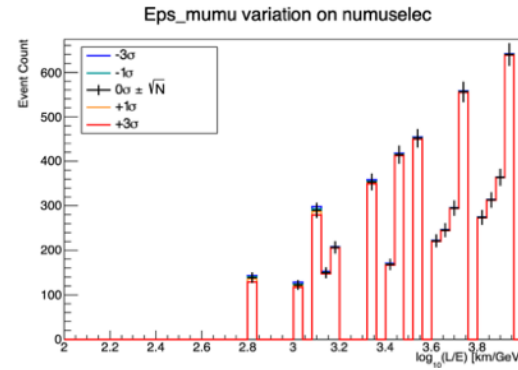
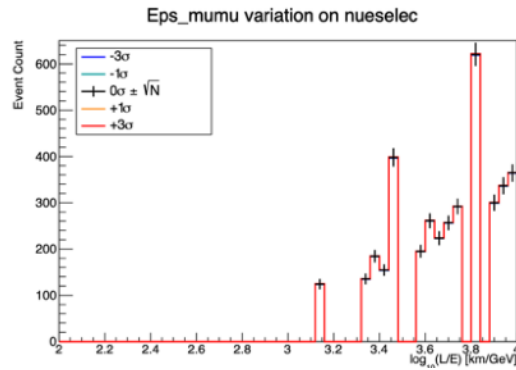


Fit error envelope with analytical function

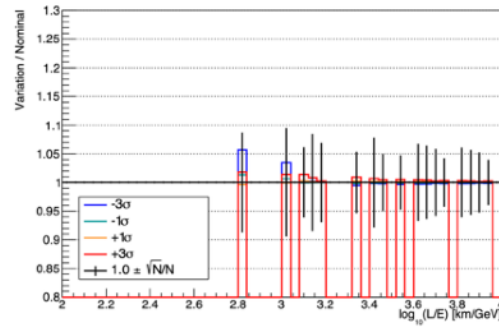
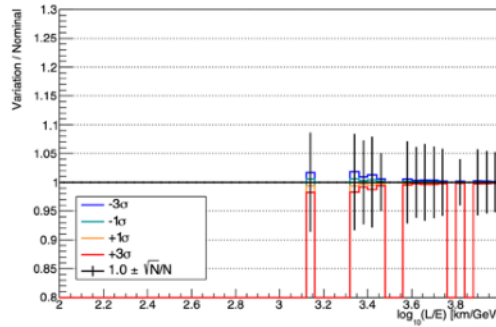
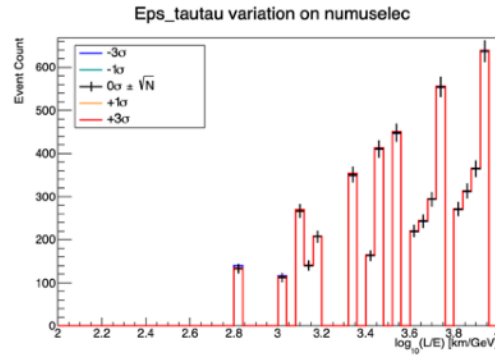
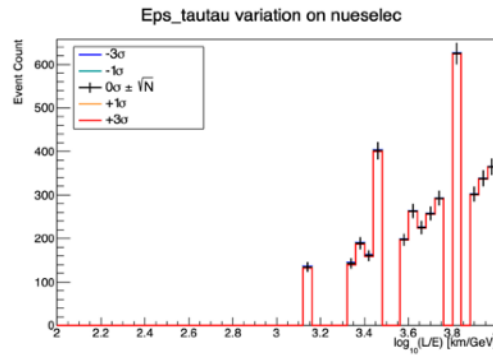
- Fitting with TSpline3 vs fitting with 3rd order polynomial
- TSpline3 goes through center of each bin, over fitting



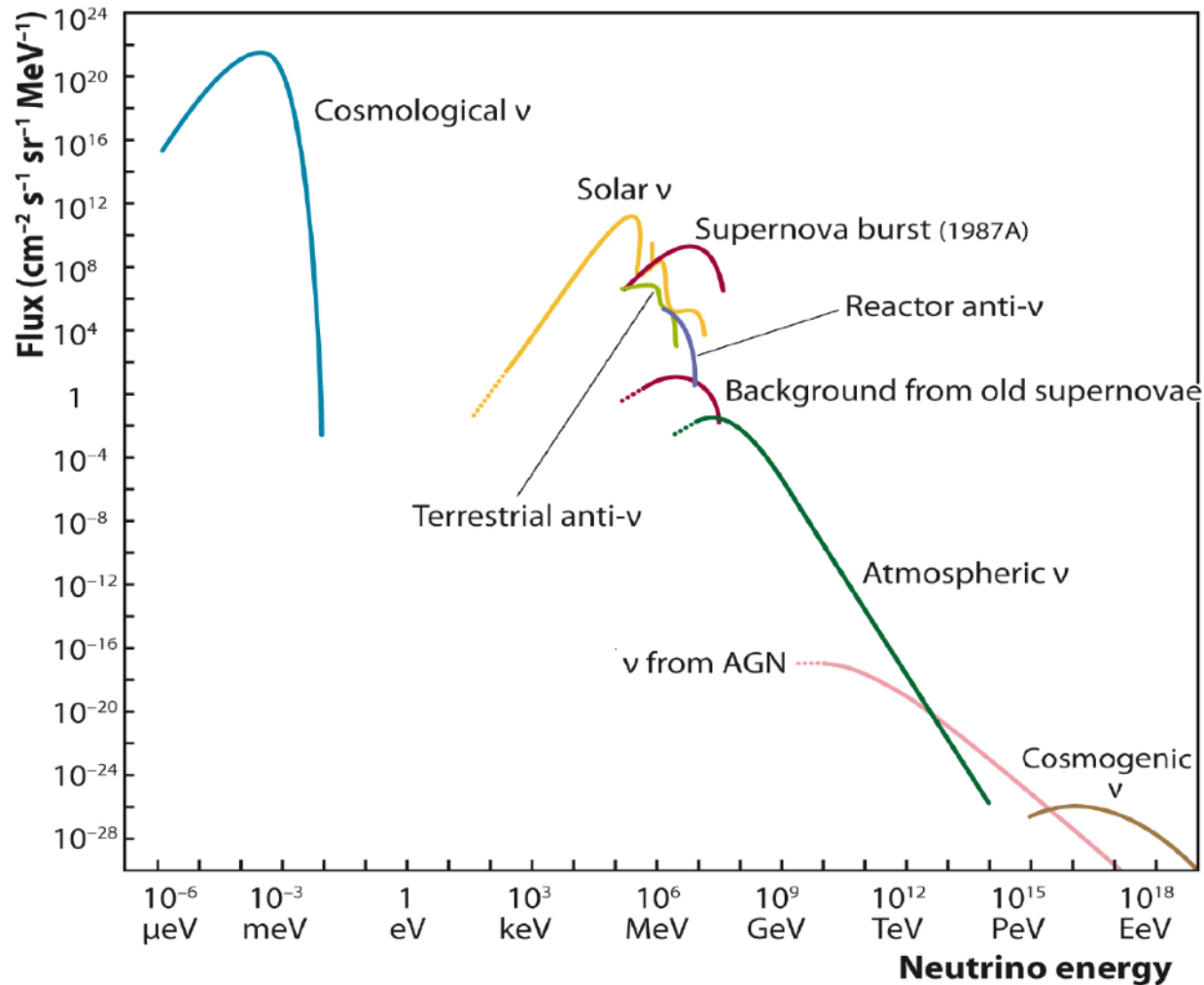
NSI set to be 0.2, 1sigma = 0.02



NSI set to be 0.2, 1sigma = 0.02



Neutrino source and energy



<https://ar5iv.labs.arxiv.org/html/1207.4952>

Bayesian Inference

- $P(\vec{\theta} | \vec{D}) = \frac{P(\vec{D} | \vec{\theta})P(\vec{\theta})}{P(\vec{D})}$
- Likelihood: probability of observing some dataset D, given some model with parameter values theta,
- $P(\vec{D}) = \int P(\vec{D} | \theta)P(\vec{\theta})d\theta$ marginal likelihood
- Difficult to evaluate marginal likelihood
- $P(\vec{\theta} | \vec{D}) \propto P(\vec{D} | \vec{\theta})P(\vec{\theta})$
- Systematics application 2 methods:
 - Covariance method: $-2\ln L_{total} = -2\ln L_{stat} + (w - 1)^T V_{flux}^{-1} (w - 1)$
 - Functional response: event by event reweighing:
 - $\Phi(E, \cos\theta) = \Phi(E, \cos\theta)[1 + f(E, \cos\theta)]$
 - Event reweighing: $w_{flux} = 1 + f(E_i, \cos\theta_i)$

Covariance Matrix

- For MCMC 2D posterior

- (θ_1, θ_2) pairs

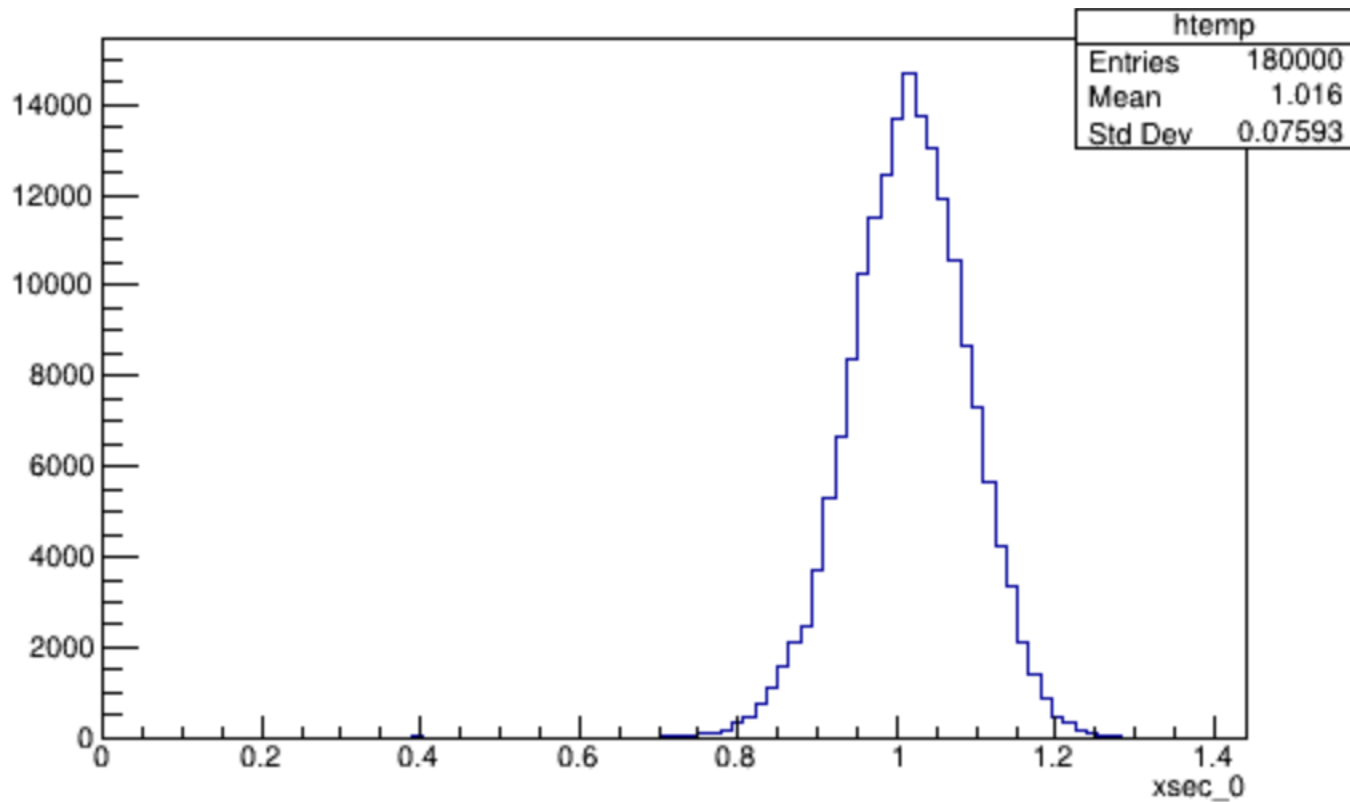
- $$\text{Cov}(\theta_1, \theta_2) = \frac{1}{N-1} \sum_{i=1}^N (\theta_1^{(i)} - \bar{\theta}_1)(\theta_2^{(i)} - \bar{\theta}_2)$$

- $$\log \mathcal{L}(\theta) = -\frac{1}{2}(\theta - \mu)^T C^{-1}(\theta - \mu) - \frac{1}{2} \log \det C + \text{const},$$
 C is posterior covariance matrix

- Gaussian probability density:

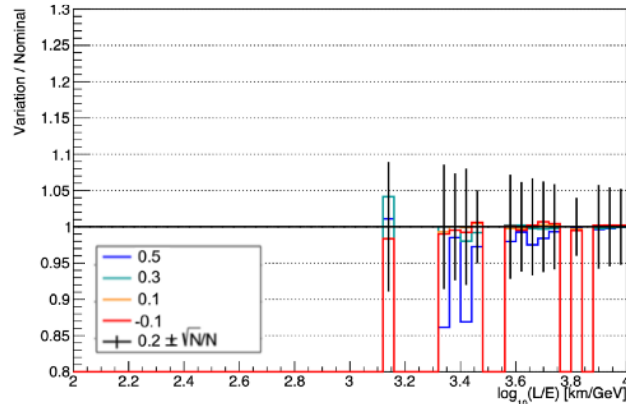
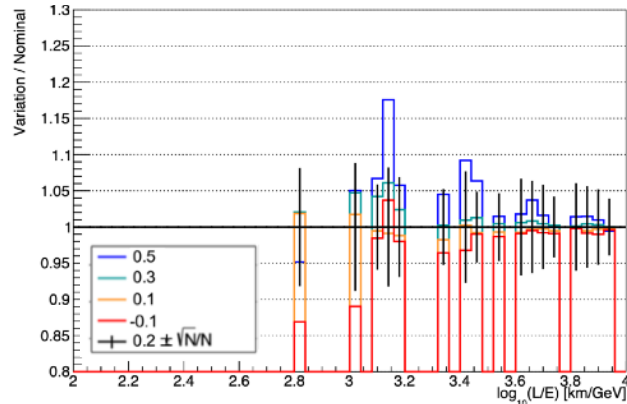
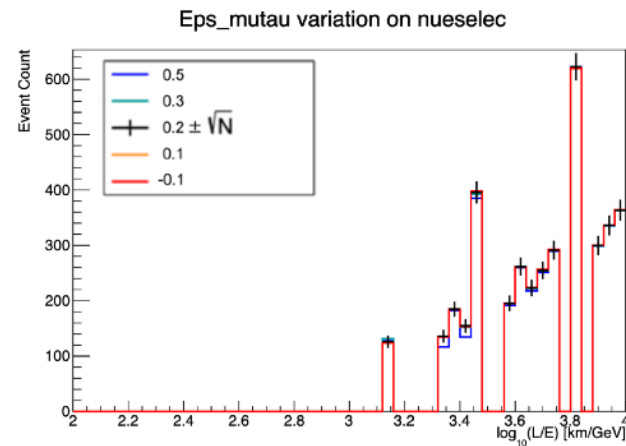
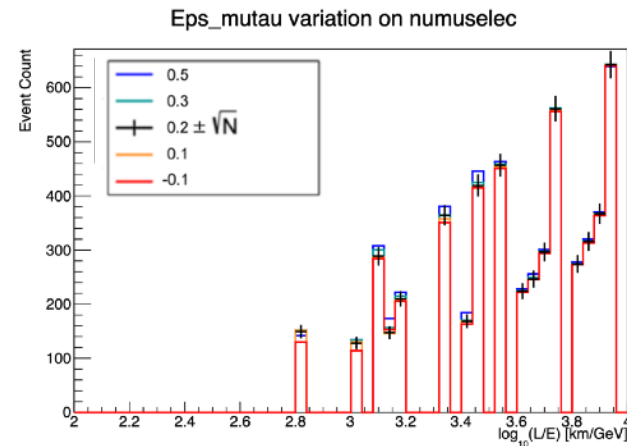
- $$p(\theta) = \frac{1}{\sqrt{(2\pi)^k \det C}} \exp\left[-\frac{1}{2}(\theta - \mu)^T C^{-1}(\theta - \mu)\right]$$

Flux systematics posterior



NSI Signal Visualization

- Example: $\varepsilon_{\mu\tau} = -0.1, 0.1, 0.2, 0.3, 0.5$
- Visible shift in L/E distribution demonstrate DUNE sensitivity



Comparison to Current Limits

- My results: (with e^- coupling)
 - $\varepsilon_{\mu\tau}$: $[-0.060, 0.229]$
 - $\varepsilon_{\mu\mu}$: $[-0.394, 0.393]$
 - $\varepsilon_{\tau\tau}$: $[-0.396, 0.382]$
 - $\varepsilon_{\tau\tau} - \varepsilon_{\mu\mu}$: $[-0.138, +0.119]$
 - Exposure: 10 years, 0.1 to 10 GeV
- Current Global NSI limits:
 - $\varepsilon_{\mu\tau}$: $[-0.0063, +0.016]$ with e^- coupling
 - $\varepsilon_{\mu\mu}$: $[-0.2, +0.1]$
 - $\varepsilon_{\tau\tau}$: $[-0.17, +0.093]$
 - $\varepsilon_{\tau\tau} - \varepsilon_{\mu\mu}$: $[-0.040, +0.047]$
 - arXiv:1805.04530 2020
- NOvA beam
 - $\varepsilon_{\mu\tau}$ not included in study
- IceCube (universal coupling)
 - $\varepsilon_{\mu\tau}$: $[-0.0041, +0.0031]$
 - 305,735 up-going ν_μ 500 GeV to 9976 GeV
- Super-K atmospheric
 - $\varepsilon_{\mu\tau}$: $[-0.046, +0.019]$
 - $\varepsilon_{\tau\tau} - \varepsilon_{\mu\mu}$: $[-0.050, +0.138]$
 - 15 years exposure
 - Mahdi Taani. “Non-Standard Neutrino Interaction Analysis with Atmospheric Neutrino Data in Super-Kamiokande I-IV and the Design of the Hyper-Kamiokande Outer Detector”